

Lab 1 -- Introduction to Electronic Instruments Lab Report

Name: Chen Yi Hui	ID: 3180300684	Major: Computer Science
Course: Digital Logic Design	Groupmates: Tang Anna, 伍美萍, Choi Justin, Malmaerg Joshua, 冯乐人	
Date: 2019-09-11	Place: ZiJinGang East 4-509	Instructor: Hong QiJun

I. Experiment Purpose and Requirements

- 1.1 Introduction to Electronic Instruments
- 1.2 Digital multimeter, oscillograph, signal generator, DC power supply.
- 1.3 Measure frequency and amplitude of sinusoidal wave with oscillograph
- 1.4 Measure sinusoidal wave edges and delay with oscillograph
- 1.5 Measure voltages, resistances and diodes with digital multimeter

II. Experiment Contents and Principles

2.1 Electronic Instruments

2.1.1



Figure 1 Circuit design experimental box

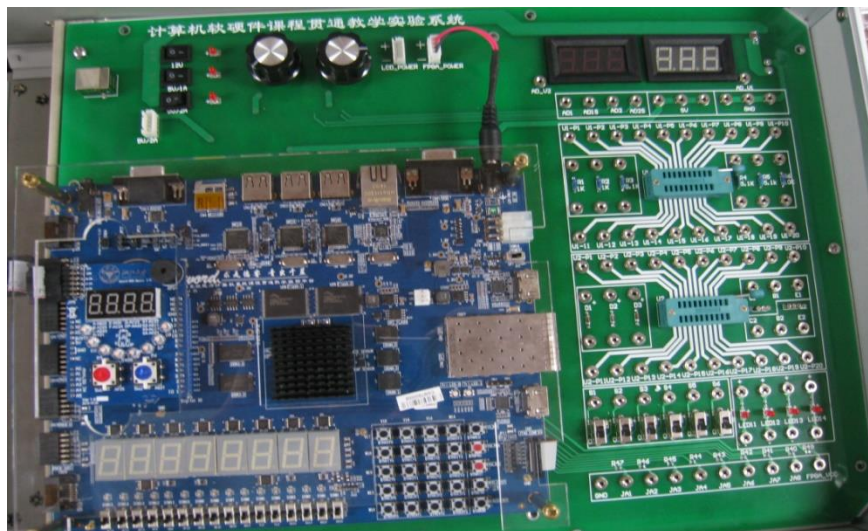


Figure 2 Circuit design board

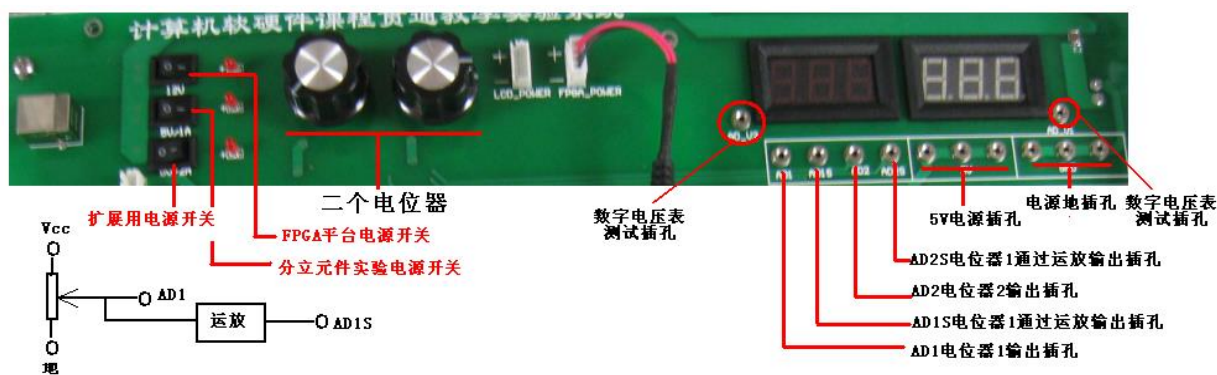


Figure 3 Circuit design board switch

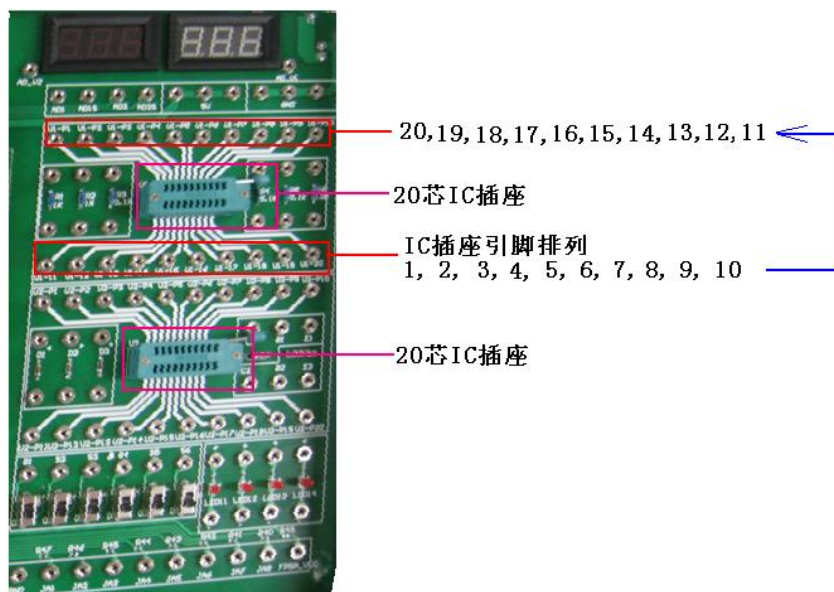


Figure 4 Circuit design board chip socket

2.1.2 Oscilloscope Panel

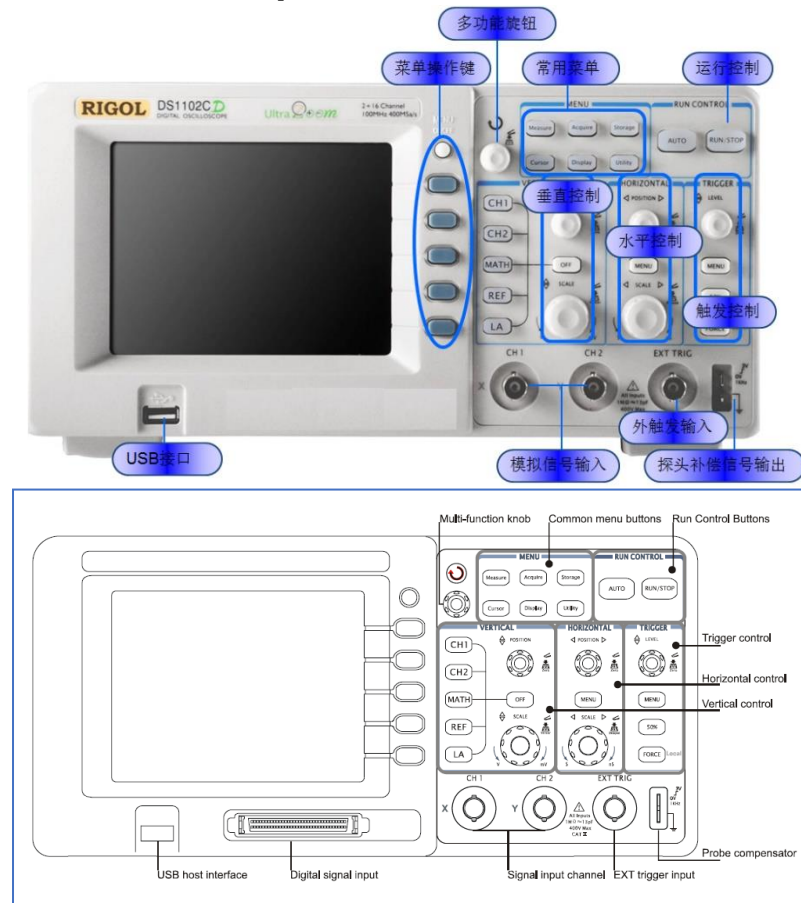


Figure 5&6 Oscilloscope RIGOL-DS162 and control panel

2.1.3 Signal input cable of oscilloscope

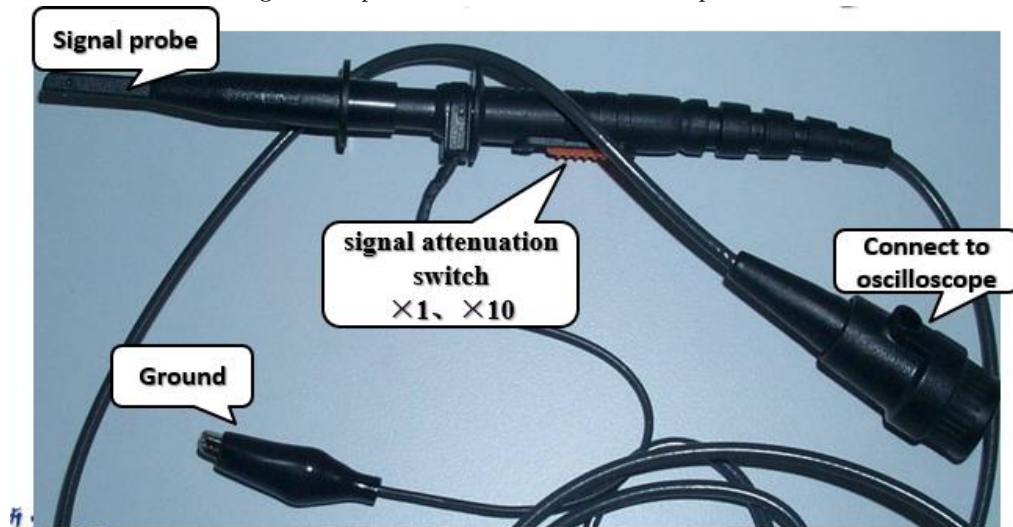


Figure 7 cable of oscilloscope

2.1.4 YB1638 Signal Generator

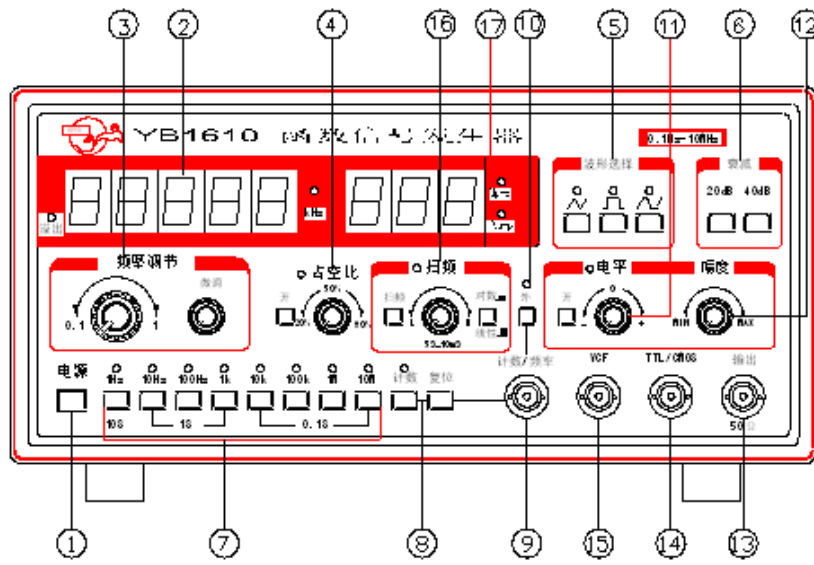


Figure 8 YB1638 Signal Generator

- 1 Power
- 2 Frequency display
- 3 Frequency adjust
- 5 Output wave option
- 7 Output frequency range adjust
- 11 Amplitude adjust
- 13 Output three types of waveform
- 14 TTL Output

2.1.5 Digital Multimeter



Figure 9 Digital Multimeter

- Measure DC voltage and current
- Measure AC voltage and current
- Measure resistance
- Test diode and triode

2.1.6

Resistor table

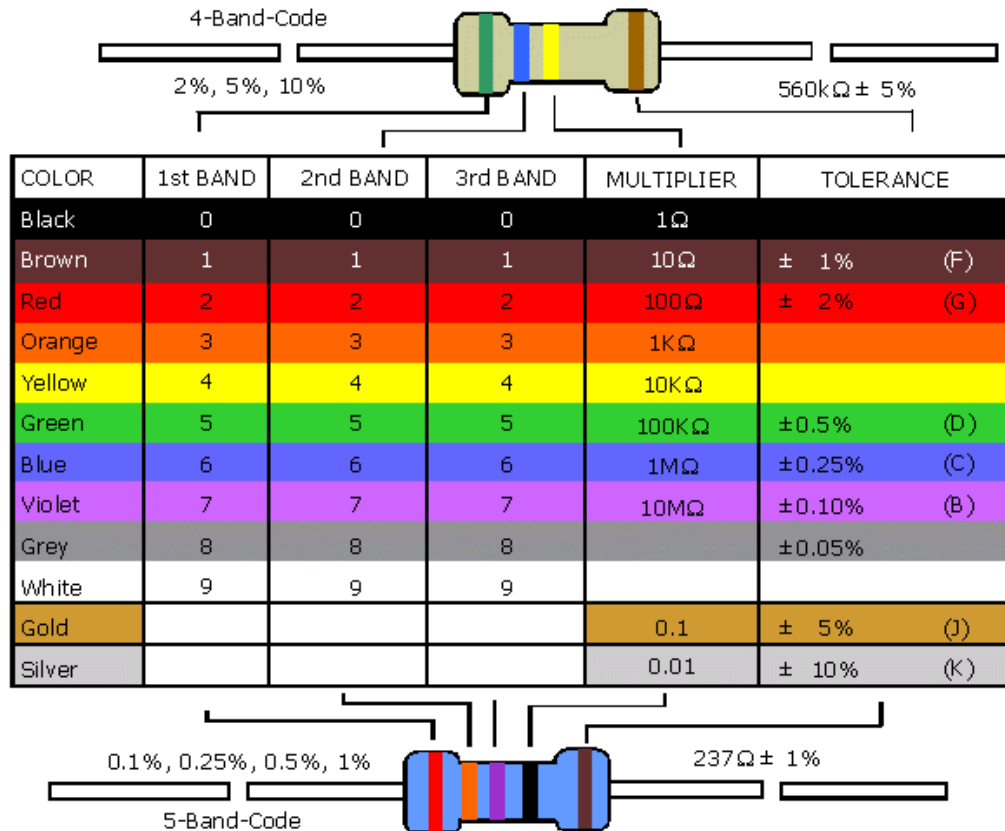


Figure 10 Resistor table

2.2 Content

2.2.1 Measure parameters of sinusoidal signal with oscilloscope

Use the oscilloscope to measure the pulse waves generated by signal generator. Generate sine wave in frequency of 100Hz, 10KHz and 100KHz, and then measure the frequency and period with oscilloscope.

Figure 11 Data table

	Sinusoidal signal	Div Count	Sensitivity	Measured value	
Amplitude	3V~5V	Div	V/Div	V	
Period/Frequency	100Hz	Div	ms/Div	ms	Hz
Amplitude	2V~10V	Div	V/Div	V	
Period/Frequency	10KHz	Div	μ s/Div	μ s	Hz

Amplitude	4V~6V	Div	V/Div	V	
Period/Frequency	100KHz	Div	μ s/Div	μ s	Hz

2.2.2 Measure the output voltage of YB1638 signal generator

- Generate 1KHz signal with effective value of voltage in range of 1V-3V. Then connect the output to oscilloscope and measure the peak-peak voltage of the signal with oscilloscope.
- Measure the AC voltage with digital multimeter.
- Calculate the effective value from the peak-peak voltage and compare it with the reading of multimeter.

Figure 12 Data table

Output frequency	oscilloscope read		Effective value	Multimeter read
1KHz	div	V/div	V	V
50hz	Div	V/div	V	V
1MHz	Div	V/div	V	V

- Connect signal generator to the multimeter, with red probe to positive and black probe to negative. The multimeter is set to AC mode with proper voltage range. Adjust the signal amplitude as 3V.
- Output the signal to the oscilloscope to measure the peak-peak voltage

$$V_{p-p}. \quad \text{Effective value} = V_{P-P} / 2\sqrt{2}$$

2.2.3 Measure DC voltage of experiment box with digital multimeter

- Put the red probe into V/Ω slot and black probe into COM slot of digital multimeter
- Set the digital multimeter to DC mode with proper range and measure the voltage of DC power of experiment box.
- Measure the voltage of all three DC outputs and fill the table.

Figure 13 Data table

DC output	oscilloscope read	sensitivity	Oscilloscope converted value	Multimeter Read
+5V	Div	V/Div	V	V

2.2.4 Test the state of diode

- Put the black probe into COM slot and read probe into V/ Ω slot. And red probe is positive.
- Set the multimeter to model “”, connect the two probes to two legs of diode. If the display of the multimeter is between 0.6~0.7, then the red probe connected leg is positive; if the display number is “1”, then the black probe connected leg is positive

III. Experiment Instruments and Materials

oscilloscope RIGOL-DS162	x1
Signal Generator YB1638	x1
Digital Multimeter	x1
Experiment Box of Logic Circuit Design	x1

IV. Experiment Procedures and Operations

4.1 Measure parameters of sinusoidal signal with oscilloscope

Connect the signal generator output cable to the input cable of the oscilloscope, and ground to ground. Turn on the signal generator and oscilloscope, turn the frequency knob to desire output frequency. Adjust the output voltage by turning the amplifier knob, press auto to display sinusoidal wave on the screen of the oscilloscope. Read and record data on the screen, then perform calculation. Test and record frequencies of 100Hz, 10kHz and 100kHz with voltage 3V-5V, 2V-10V and 4V-6V respectively. And fill in diagram.11 data table.

4.2 Measure the output voltage of YB1638 signal generator

Generate frequency signal with voltage 4V to 6V on the YB1638 signal generator. Read and record the effective voltage from the oscilloscope using measure feature or calculate effective voltage with given voltage input using the effective voltage formula: $Effective\ value = V_{p-p} / 2\sqrt{2}$. Use the digital multimeter connecting its signal and ground needle with the

connection of signal generator and oscilloscope. Turn the digital multimeter to AC 20V and the digital multimeter will display the effective voltage. Record data on diagram 12.

- 4.3 Measure the DC voltage of the experiment box with digital multimeter
Use a digital multimeter with needles properly attached, red needle attaches to the VΩmA, and black needle attaches to the COM.

Turn the digital multimeter to DC mode with proper measuring range and measure the voltage of DC power of the experiment box. Put red needle into the 5V power source and black needle into ground in the experiment box. Add oscillograph in the connection of digital multimeter and experiment box, and press “AUTO”. Record data from the digital multimeter and the oscillograph onto data table of diagram 13.

- 4.4 Test state of diode
Use a digital multimeter with needles properly attached, red needle attaches to the VΩmA, and black needle attaches to the COM.

Turn the knob on the digital multimeter to , connect the two needles with the diode on the circuit board in the experiment box. Digital multimeter will display a number from 0.6 to less than 1 if the red needle is connected to positive, multimeter will display number “1” if the black needle is connected to positive.

V. Experiment Result and Analysis

- 5.1 Measure parameters of sinusoidal signal with oscillograph

Figure 14 Sinusoidal signal with oscillograph data table

	Sinusoidal signal	Div Count	Sensitivity	Measured value	
Amplitude	3V~5V	5 Div	1V/Div	5V	
Period/Frequency	100Hz	2 Div	5ms/Div	10ms	100Hz
Amplitude	2V~10V	3 Div	1V/Div	3V	
Period/Frequency	10KHz	2Div	50 μ s/Div	100 μ s	10kHz
Amplitude	4V~6V	5Div	1V/Div	5V	
Period/Frequency	100KHz	2Div	50 μ s/Div	10 μ s	100kHz

Analysis: Sinusoidal wave on the oscilloscope can be measured by calculating the divisions and values of each division. Or we can use the “measure” feature on the oscilloscope control panel to display frequency and voltage info. Signal output from the signal generator matches readings on the oscilloscope, oscilloscope is a reliable instrument to use when measuring sinusoidal to prevent human error, but proper setup procedure is required.

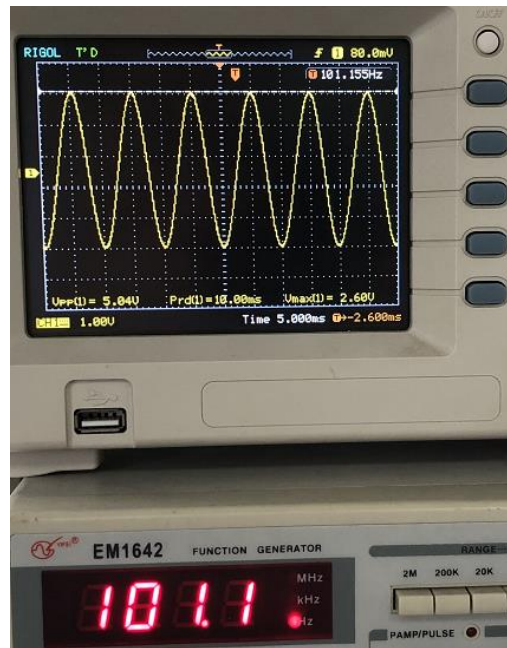


Figure 15. Measurement with 5V and 101.1Hz

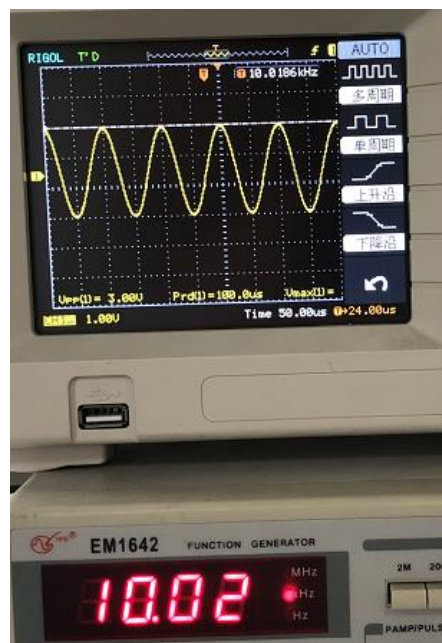


Figure 16. Measurement with 3V and 10.02Hz

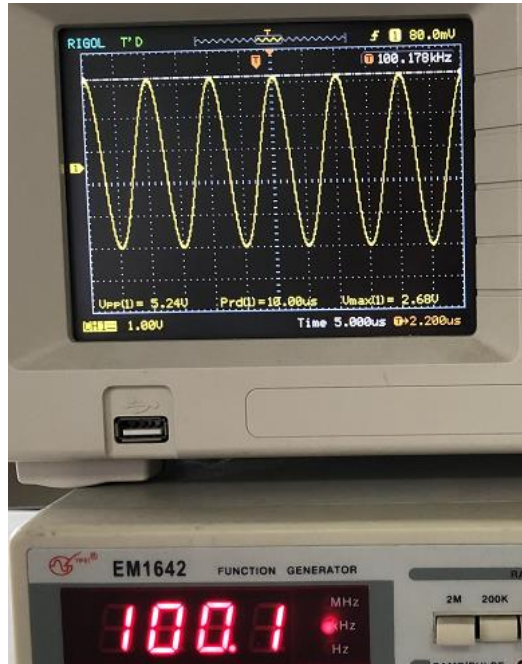


Figure 17. Measurement with 3V and 10.02Hz

5.2 Measure the output voltage of YB1638

Figure 18. YB1638 output data table

Output frequency	oscilloscope read		Effective value	Multimeter read
1KHz	5 Div	1V/div	1.77V	1.68V
50hz	5 Div	1V/div	1.76V	1.72V
1MHz	5 Div	1V/div	1.77V	0V

Analysis: Read V_{p-p} from the oscillograph and calculate the effective voltage value

using the formula $\text{Effective value} = V_{P-P} / 2\sqrt{2}$, or simply use the measure feature on the oscillograph and record the effective value. It shows that there is difference from what the oscilloscope shows and what the multimeter reads. It is unclear what might have caused this problem. When comparing data with different frequency we can conclude that multimeter readings are less than oscilloscope readings.

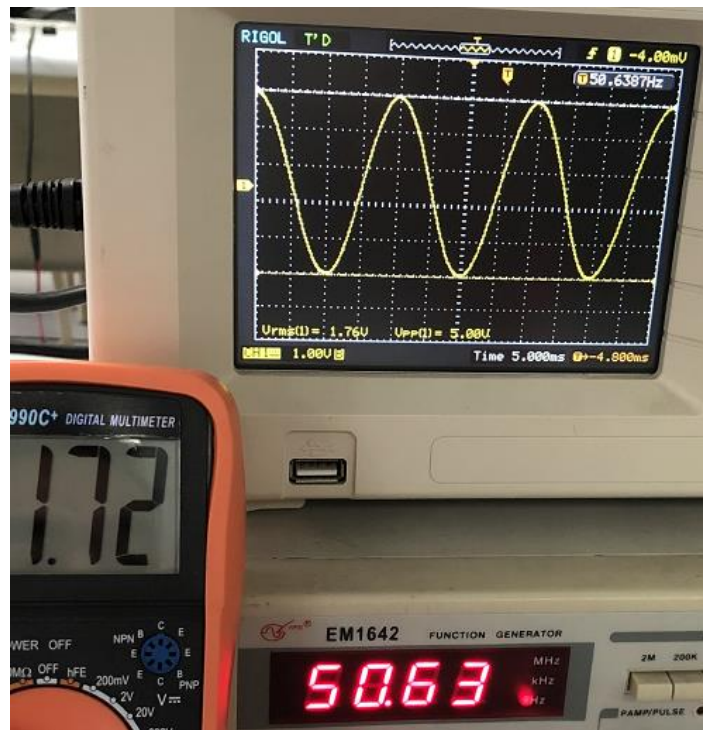


Figure 19. Measurement 50Hz 5Vp-p

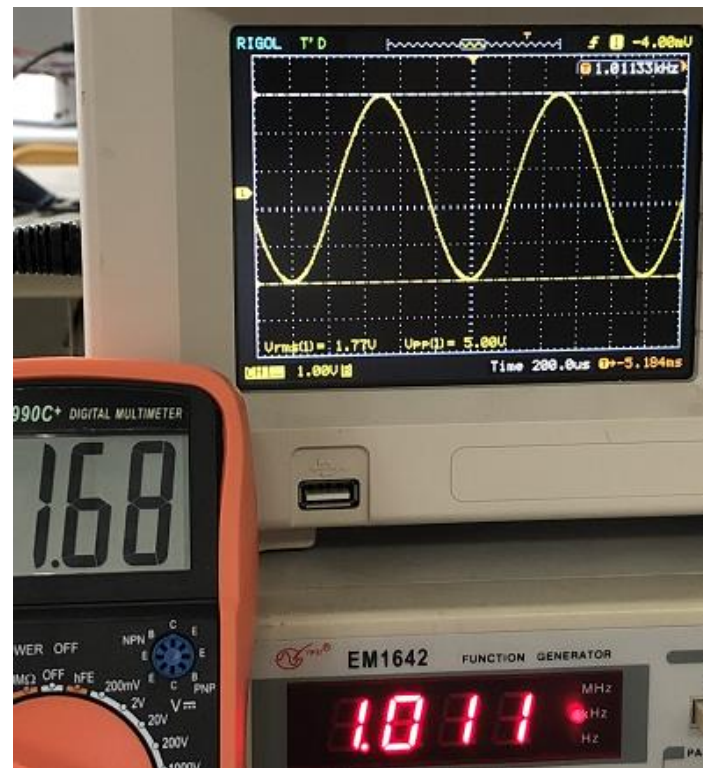


Figure 20. Measurement 1kHz 5Vp-p

5.3 Measure DC voltage of the experiment box with digital multimeter

Figure 21. Experiment box DC voltage data table

DC output	oscilloscope read	sensitivity	Oscilloscope converted value	Multimeter Read
+5V	5.2Div	1V/Div	5.19V	4.97V

Like the result of measuring YB1638 signal generator, multimeter reading seems to be off and displays value less than what the oscilloscope displays.

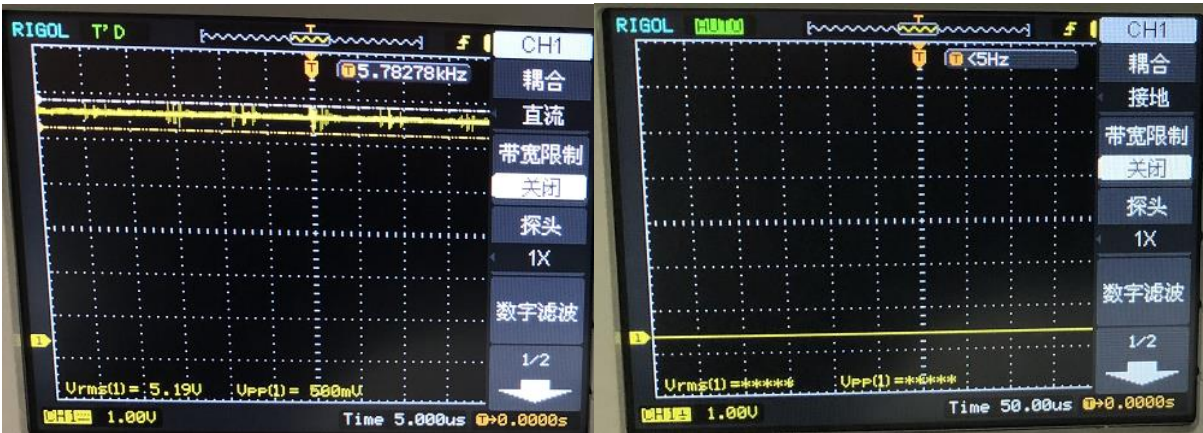
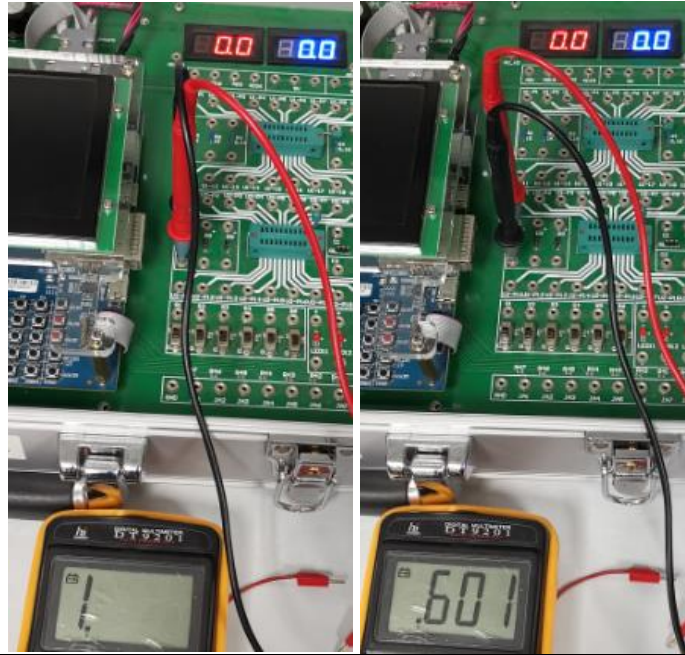


Figure 22. On the left shows when switch for DC is closed. On the right shows when switch for DC is open(ground).



Figure 23. Value on the digital multimeter

5.4 Test state of diode



	Value:
Diode forward:	0.592
Diode reverse:	1

Figure 24. Diode forward

Analysis: Data recorded is accurate according to the instructions given.

VI. Discussion and Revision

This experiment introduced basic electronic instruments in detail and allowed us to work with them. Performing operations on the electronic instruments gained hand-on experience, very confident If I am to use these instruments in the future. However, it was my first time using these instruments, it was challenging to follow the procedures.

Lab 2 -- Fundamental Logic Gate Circuit

Lab Report

Name: Chen Yi Hui	ID: 3180300684	Major: Computer Science
Course: Digital Logic Design	Groupmates: Tang Anna, 伍美萍, Choi Justin, Malmaerg Joshua, 冯乐人	
Date:2019-09-18	Place: ZiJinGang East 4-509	Instructor: Hong QiJun

I. Experiment Purpose and Requirements

- 1.1 Understand Basic Structure of logic circuit
- 1.2 Understand On/Off states of Diode
- 1.3 Learn how to design a simple logic gate circuit with diodes and triodes
- 1.4 Learn the structure of the simplest logic gate circuit

II. Experiment Contents and Principles

2.1 Experiment contents

- 2.1.1 Design AND gate circuit with diodes, and measure the input and output voltage, and give the logic function table.
- 2.1.2 Design OR gate circuit with diodes, and measure the input and output voltage, and give the logic function table.
- 2.1.3 Design NOT gate circuit using reverse characteristic of triode, and measure the input and output voltage, and give the logic function table.
- 2.1.4 Design NAND gate circuit with previous circuit, and measure the input and output voltage, and give the logic function table.

2.2 Experiment principle

- 2.2.1 Common Logic voltage level

	V_{CC}/V	V_{OH}/V	V_{OL}/V	V_{IH}/V	V_{IL}/V	explain
TTL	5.0	≥ 2.4	≤ 0.4	≥ 2.0	≤ 0.8	Floating input is High level
LVTTL	3.3	≥ 2.4	≤ 0.4	≥ 2.0	≤ 0.8	
LVTTL	2.5	≥ 2.0	≤ 0.2	≥ 1.7	≤ 0.7	
CMOS	5.0	≥ 4.45	≤ 0.5	≥ 3.5	≤ 1.5	Input Resistor Is very big
LVC MOS	3.3	≥ 3.2	≤ 0.1	$\geq 2.0V$	≤ 0.7	
LVC MOS	2.5	≥ 2.0	≤ 0.1	≥ 1.7	≤ 0.7	
RS232	$\pm 12 \sim 15$	$-3 \sim -15$	$3 \sim 15$	$-3 \sim -15$	$3 \sim 15$	Negative logic

Figure 1 Common Logic voltage level
Different voltage level for different circuits and conductors

2.2.2 AND gate circuit with diode

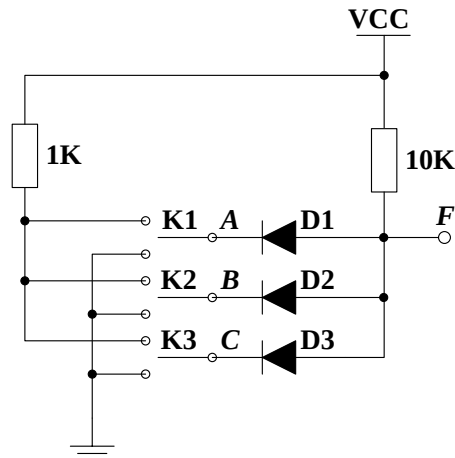


Figure 2 AND gate circuit

When one of A, B and C is Low level, F is Low level. When All A, B and C are High Level, F is High Level.

$$F=ABC$$

<i>C</i>	<i>B</i>	<i>A</i>	<i>F</i>
L	L	L	L
L	L	H	L
L	H	L	L
L	H	H	L
H	L	L	L
H	L	H	L
H	H	L	L
H	H	H	H

Figure 3 Truth table for AND gate circuit

2.2.3 OR gate circuit with diode

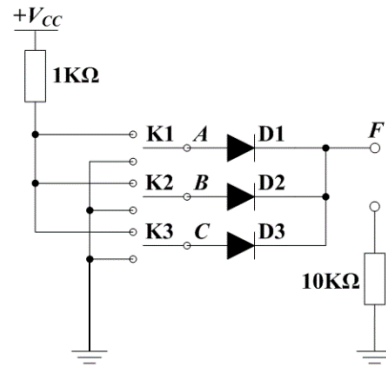


Figure 4 OR gate circuit

When one of A, B and C is High level, F is High level. When All of A, B and C are Low Level, F is Low Level.

OR gate principle: $F = A+B+C$

<i>C</i>	<i>B</i>	<i>A</i>	<i>F</i>
L	L	L	L
L	L	H	H
L	H	L	H
L	H	H	H
H	L	L	H
H	L	H	H
H	H	L	H
H	H	H	H

Figure 5 Truth table for OR gate circuit

2.2.4 NOT gate with triode

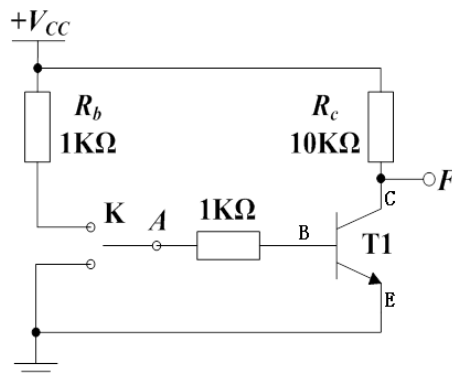


Figure 6 NOT gate circuit

When A is High Level, triode T1 is in Saturation status, $V_{CE} \approx 0.3V$, then F is low level. When A is low level, $I_B = 0$, T1 is in cut-off status, then $I_C = 0$, F is High level.

$F = \text{NOT } A$

<i>A</i>	<i>F</i>
L	H
H	L

Figure 7 Truth table for NOT gate circuit

2.2.5 NAND gate Circuit with diode and triode

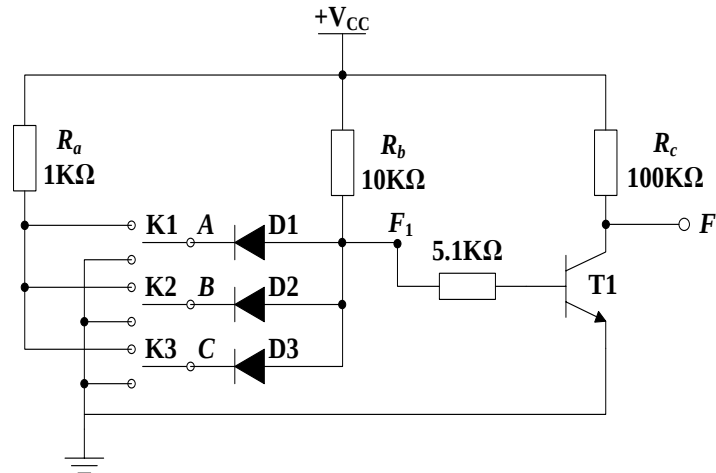


Figure 8 NAND gate circuit

When one of A, B, C are low level, T1 is in cut-off status, and F is high level. When all of A, B, C is high level, T1 is in saturation status, and F is low level.

$$F = \text{NOT}(ABC)$$

<i>C</i>	<i>B</i>	<i>A</i>	<i>F</i>
L	L	L	H
L	L	H	H
L	H	L	H
L	H	H	H
H	L	L	H
H	L	H	H
H	H	L	H
H	H	H	L

Figure 9 Truth table for NAND gate circuit

2.2.6 DC switches

We can use switches of S1-S6 built-in in the experiment box instead of VCC and control the input to circuits.

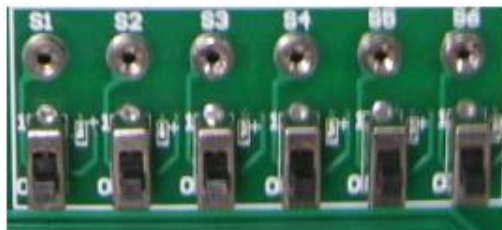


Figure 10 DC Switches in experiment box

The circuit to produce high level and low level of voltage is connected in experiment box.

with the switches of S1/S2/S3/S4/S5/S6, each jack produces input high level and low level of voltage

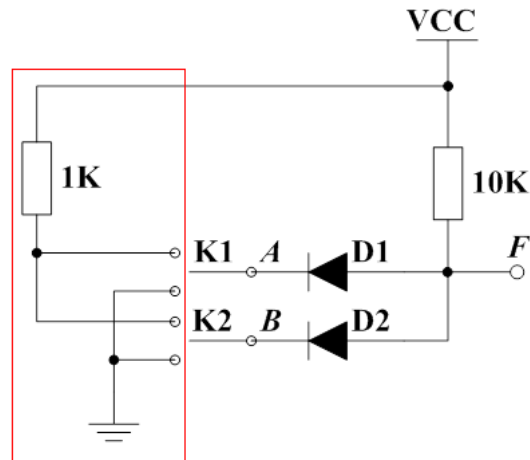


Figure 11 Area within red rectangle can be replace by DC switches

III. Experiment Instruments and Materials

3.1 Instruments

Oscilloscope RIGOL-DS162	1
Signal Generator YB1638	1
Digital Multimeter	1
Experiment Box of Logic Circuit Design	1

3.2 Materials

Diode IN4001	3
Triode 9013	1
LED	1
Resistor:	
• 10K Ω	5
• 5.1K Ω	3
• 1K Ω	5

IV. Experiment Procedures and Operations

4.1 Design AND gate circuit with diode

- Connect wires in the experiment box according to AND gate circuit figure.

- Vcc is connected to +5V DC in experiment box.
- Switches S1/S2/S3/S4/S5/S6 are used to control the voltage level of A and B.
- Measure the output F, and check the relationship: $F = AB$

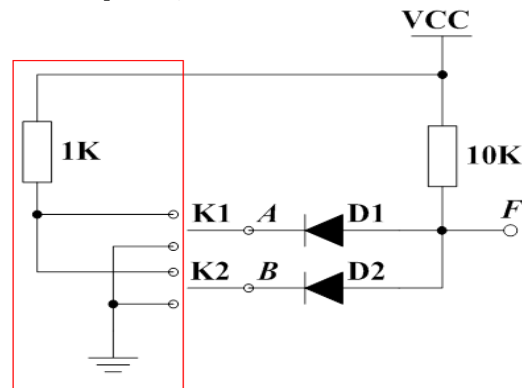


Figure 12 AND gate circuit

4.2 Design OR gate circuit with diode

- Connect the circuit according to the OR gate circuit figure.
- Switches S1/S2/S3/S4/S5/S6 are used to control the voltage level of A and B.
- Measure the output F and check the relationship: $F = A+B$

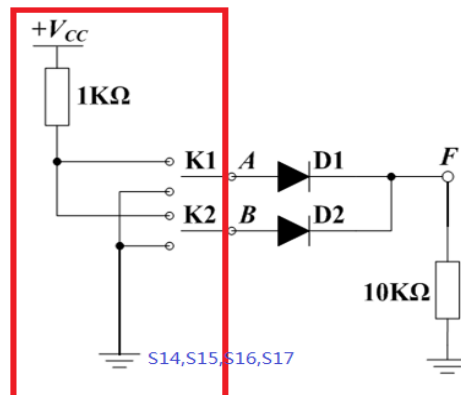


Figure 13 OR gate circuit

4.3 Design NOT gate circuit with triode

- Connect the circuit according to the NOT gate circuit figure.
- Vcc is connected to +5V DC in experiment box.
- Switches S1/S2/S3/S4/S5/S6 are used to control the voltage level of A and B.
- Measure the output and check the relationship: $F = \text{NOT}(A)$

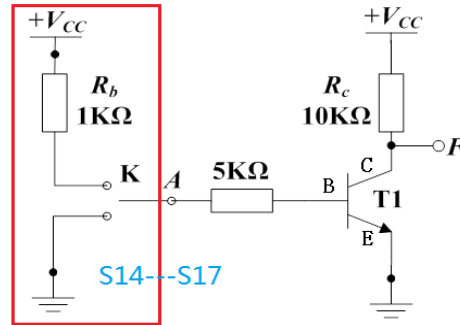


Figure 14 NOT gate circuit

4.4 Design NAND gate circuit with triode and diode

- Connect the circuit according to the NAND gate circuit figure.
- Vcc connects to +5V DC in experiment box.
- Input high level and low level into A and B with the switches of S1/S2/S3/S4/S5/S6.
- Measure the output F with Digital Multimeter, Check the relationship: $F = \text{NOT}(AB)$

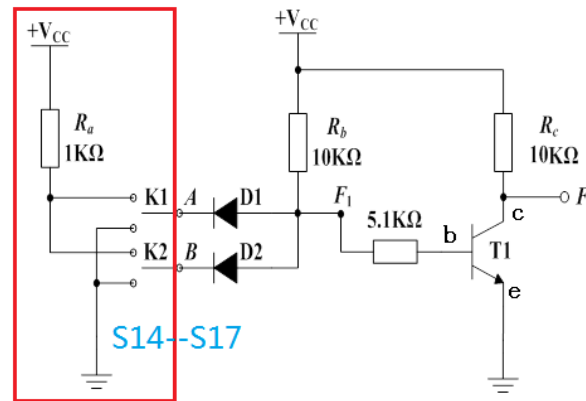


Figure 15 NAND gate circuit

4.5 Test the polarity of triode

- Plug the red pen into VΩmA slot, plug the black pen into COM slot, find whether the triode is PNP or NPN type, find the basic pin “b”.
- Switch multimeter in hFE position, place Basic pin in b slot. Place the other two pin in c or e, read the value.
- Then replace c and e and Read the value again.
- Record the c and e while the value is bigger.

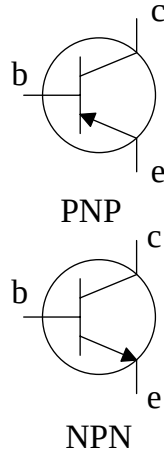


Figure 16 Polarity of triode

V. Experiment Result and Analysis

5.1 AND gate circuit with diode

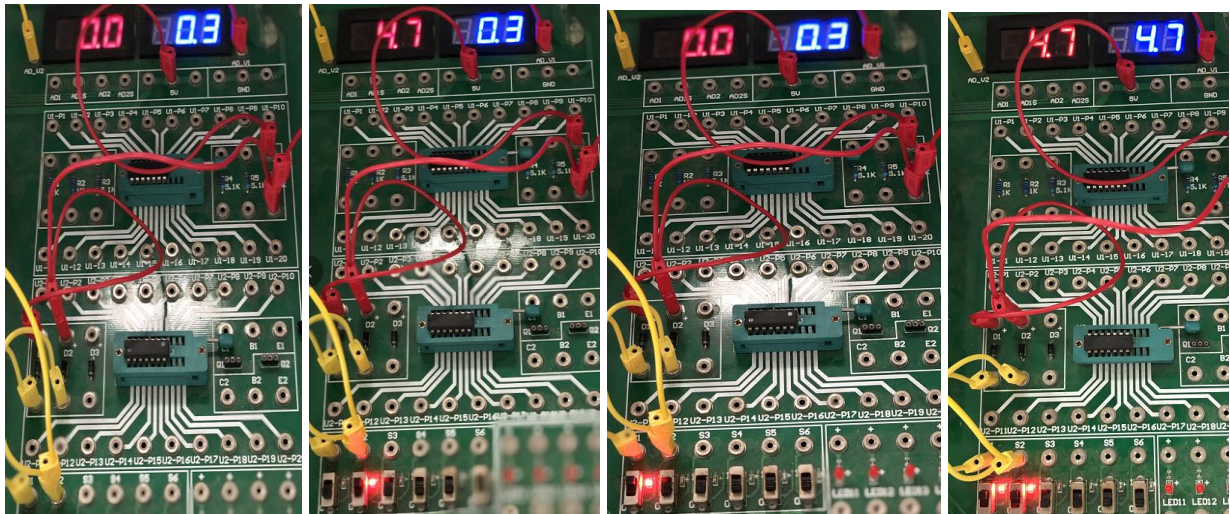


Figure 17-20 Experiment results with 10k resistor

Figure 17 A=0, b=0, F=0.3

Figure 18 A=0, B=4.7, F=0.3

Figure 19 A=4.7, B=0, F=0.3

Figure 20 A=4.7, B=4.7, F=4.7

V_A/V	V_B/V	V_F/V	F 逻辑值
0	0	0.3	L
0	4.8	0.3	L
4.8	0	0.3	L
4.8	4.8	4.7	H

Figure 21 Truth table for AND gate

Analysis: From observation, output is high when both A and B are high, low otherwise.

5.2 OR gate circuit with diode

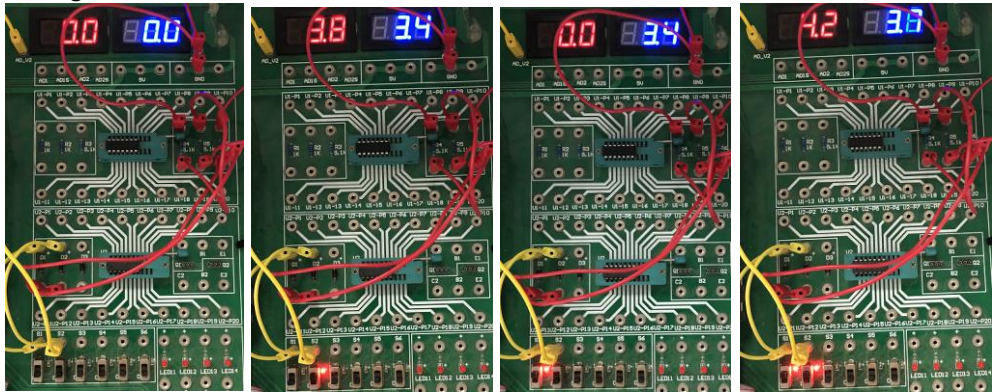


Figure 21-24 Experiment results with 20k resistor

- Figure 21 A=0, b=0, F=0
- Figure 22 A=0, B=4.7, F=4.2
- Figure 23 A=4.7, B=0, F=4.2
- Figure 24 A=4.7, B=4.7, F=4.7

V_A/V	V_B/V	V_F/V	F 逻辑值
0	0	0	0
0	1	1	1
1	0	1	1
1	1	1	1

Figure 25 Truth table for OR gate

Analysis: From the experiment, we observed that output is high when either A or B is high, and output is low only when both A and B are low.

5.3 NOT gate circuit with triode

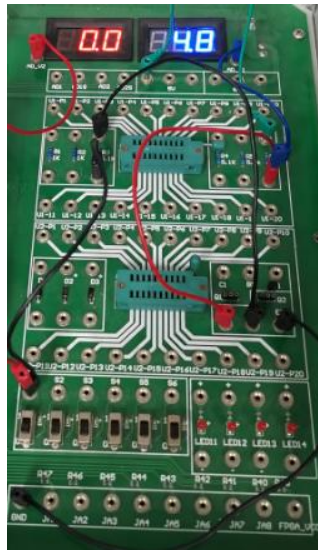


Figure 26 NOT gate circuit

V_A/V	V_F/V	F 逻辑值
0	4.8	H
2.5	0	L

Figure 27 Truth table for NOT gate

Analysis: Circuit gate inverts input.

5.4 NAND gate circuit with triode

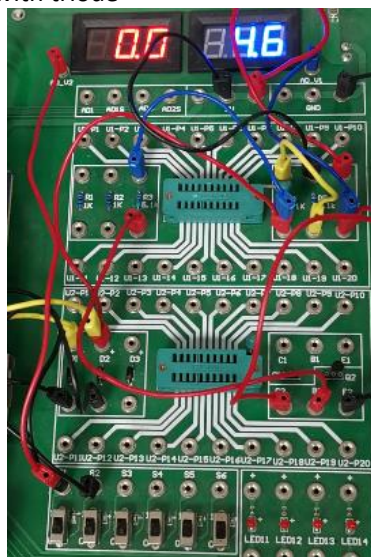


Figure 28 NAND gate circuit with 30k resistor

V_A/V	V_B/V	V_F/V	F 逻辑值
0	0	4.3	H
0	4.8	3.3	H
4.8	0	3.3	H
4.8	4.8	0	L

Figure 29 NAND gate truth table

Analysis: F is low when both A and B are high. F is high when either A or B is low.

VI. Discussion and Revision

In this experiment, I got familiar with AND, OR, NOT and NAND gate, and found ways to define them. I got to know how to build them and their structure, as well the structure of the experiment box. Understood more about diode and triode, and the use of the them. Observed the fact that the use of diode is inverted for AND gate and OR gate. Also, only triode is involved in building NOT gate and both triode and diode are involved in building NAND gate.

Lab 3 -- Function and Parameter Test for Integspecified Logic Gate Circuit Lab Report

Name: Chen Yi Hui	ID: 3180300684	Major: Computer Science
Course: Digital Logic Design	Groupmates: Tang Anna, 伍美萍, Choi Justin, Malmaerg Joshua, 冯乐人	
Date:2019-09-25	Place: ZiJinGang East 4-509	Instructor: Hong QiJun

I. Experiment Purpose and Requirements

- 1.1 Understand the functional characteristic, external electronic characteristic of basic logic gate circuit and its special usage. Digital multimeter, oscillograph, signal generator, DC power supply.
- 1.2 Understand the package and pin out of TTL NAND gate and MOS NOR gate.
- 1.3 Master the method of testing the parameters and static characteristic, and to better understand the meanings of these parameters.
- 1.4 Understand gate propagation delay.
- 1.5 Master the usage of Oscilloscope and signal generator.

II. Experiment Contents and Principles

2.1 Contents

- 2.1.1 Verify logic function of NAND gate with 74LS00.
- 2.1.2 Verify logic function of NOR gate with CD4001.
- 2.1.3 Measure t_{pd} for 74LS00.
- 2.1.4 Measure t_{pd} for CD4001.
- 2.1.5 Measure the voltage transfer characteristic and on-level and off-level V_{ON} and V_{OFF} for 74LS00.

2.2 Principles

2.2.1 Fan-out N_o

2.2.1.1 74LS00 NAND gate circuit

- Fan-Out is a parameter measure the output loads of a digital components, represents the output capacity of a component. A

logic component is a binary quantization device. the output capacity can be represented by the number of the same type of logic gates driven.

- Under the specified voltage, the maximum number of the same type of logic gates driven by the components is defined as Fan-Out.

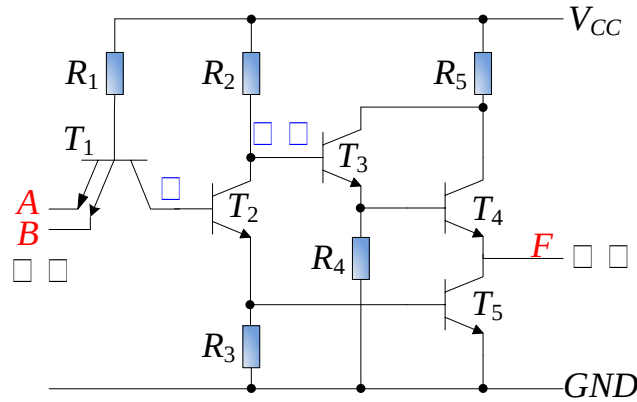


Figure 1 NAND gate circuit

- When Input A and B are at high level, T1 is off, current is small
- When Input A or B are at low level, T1 is on, current is big

2.2.2

Output High-Level V_{OH}

- Output Voltage of High-Level V_{OH} is the value when the output voltage is at high-level, which is more than 2.4V.
- V_{OH} of 74LS00 is the output voltage when input voltage is at low-level, and output current is 400uA.

2.2.3

Output Low-Level V_{OL}

- Output Voltage of Low-Level V_{OL} is the value when the output voltage is at low-level, which is less than 0.4V.
- V_{OL} of 74LS00 is the output voltage when input voltage is at high-level, and the input current is 4mA.

2.2.4

Voltage Transfer Characteristic

- The characteristic that the output voltage varies according to the input voltage.
- It sufficiently demonstrates the logical characteristics of gate input and output, and reflects the continuity of the binary quantization and the transition.

2.2.5

Off-Level V_{OFF}

- When the output voltage decreases to the lower margin of

high-level range (2.4V), the maximum input voltage is V_{OFF} . After the input voltage is larger than V_{OFF} , the output voltage will drop.

2.2.6 On-Level V_{ON}

- when the output voltage is decreased to the highest low-level (0.4V), the input voltage is V_{ON} . After the input voltage is over V_{ON} , the output voltage still decreases, slowly.

2.2.7 Noise margin

- The maximum external noise voltage imposed on the normal input that will not cause an unexpected change of the output.
- Suppose the specified high-level input voltage is V_{SH} and specified low-level input voltage is V_{SL} , then

- Low-level Noise Margin

$$V_{nL} = V_{OFF} - V_{SL} = V_{OFF} - 0.4 (V_{OL} \leq 0.4)$$

- High-level Noise Margin

$$V_{nH} = V_{SH} - V_{ON} = 2.4 - V_{ON} (V_{OH} \geq 2.4)$$

2.2.8 Average propagation delay t_{pd}

- Propagation Delay is a dynamic parameter, it is the time required for a change in value of signal to propagate from input to output. It is caused by charge and discharge of the PN junction, capacitance etc.
- Average propagation delay t_{pd} has two parts
 - delay from high-level to low-level t_{PHL}
 - delay from low-level to high-level t_{PLH}

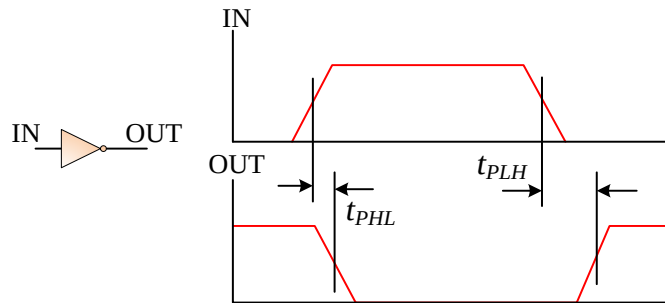


Figure 2 Measure of propagation delay

2.2.9 Current at low-level input voltage I_{iL}

- Low-level input Current I_{iL} refers to the current which flows through this input when the input is connected to GND, also

known as the short circuit input current, it can measure resistance characteristic of the low level input.

- In 74LS00 circuit, one of input is connected to GND, and the other floats, the current passing through is I_{iL} .

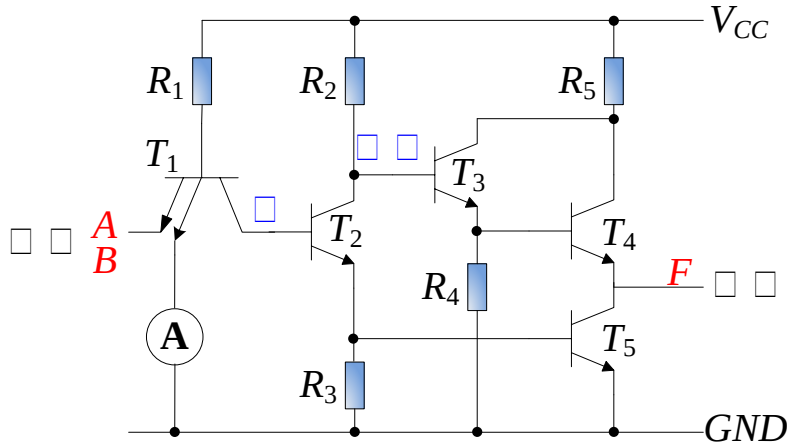


Figure 3 sample current I_{iL}

2.2.10

Current at high-level input voltage I_{iH}

- High-level Input Current I_{iH} refers to the current which flows through this input when the input is connected to high level voltage, it can measure resistance characteristic of the high level input.
- High-level Input Current I_{iH} refers to the current which flows through this input when the input is connected to high level voltage, it can measure resistance characteristic of the high level input.

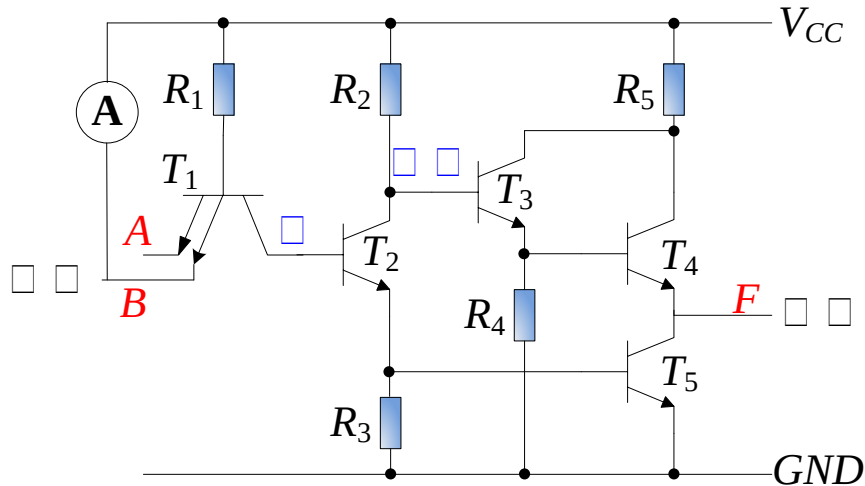


Figure 3 sample current I_{iH}

2.2.11

No-load On-State Power Consumption P_{ON}

- Empty-load On-State Power Consumption (P_{ON}) is the Power Consumption when the output is at low-level and drives no load.
- In 74LS00 circuit, P_{ON} is the power consumption when the input is at high-level and the output is at low-level without any load.

2.2.12 No-load Off-State Power Consumption P_{OFF}

- Empty-load Off-State Power Consumption (P_{OFF}) is the Power Consumption when the output is at high-level without any load.
- In 74LS00 circuit, it is the power consumption when the inputs are at low-level and the output is at high-level without any load.

III. Experiment Instruments and Materials

3.1	Experiment instruments	
	Oscilloscope RIGOL-DS162	1
	Signal Generator YB1638	1
	Digital Multimeter	1
	Experiment Box of Logic Circuit Design	1
3.2	Experiment materials	
	NAND gate with two inputs 74LS00	1
	NOR gate with two inputs CD4001	1
	Resistor	
	4.7K Ω potentiometer	1
	100 Ω /1KW	

IV. Experiment Procedures and Operations

- 4.1 Verify logic function of NAND gate with 74LS00.

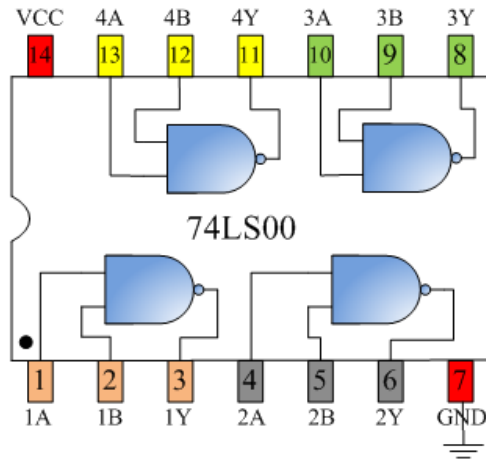


Figure 4 NAND gate with 74LS00

- Put the IC 74LS00 into socket of the Experiment Box. Pay attention to the direction of the chip.
- Connected circuit according to the diagram, port 14 is connected to V_{cc} and port 7 is to ground.
- Switches S14/S15/S16/S17 are used to generate low/high-level voltage.
- According to the truth table, measure the voltage of input A, B and output F

4.2 Verify logic function of NOR gate with CD4001.

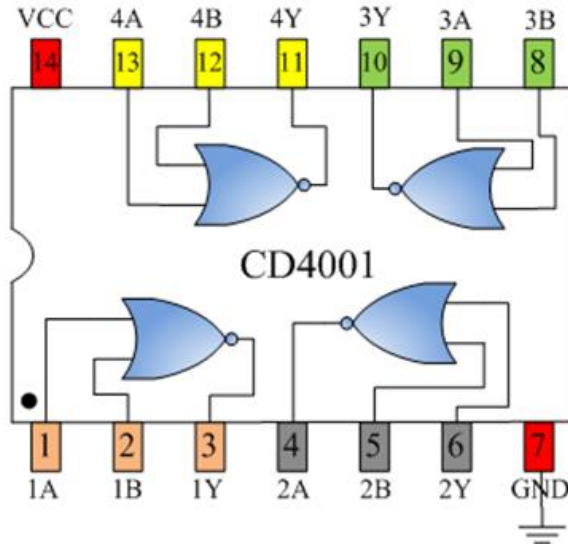


Figure 5 NOR gate with CD4001

- Put the IC into socket of the Experiment Box.
- Connected circuit according to the diagram, port 14 is connected to V_{cc} and port 7 is to ground.
- Switches S14/S15/S16/S17 are used to generate low/high-level voltage.
- According to the truth table, measure the voltage of input

A, B and the output F.

- repeat step 3~4, test the logical relationship of the other 3 gates and judge if they are ok or not.

4.3 Measure t_{pd} for 74LS00.

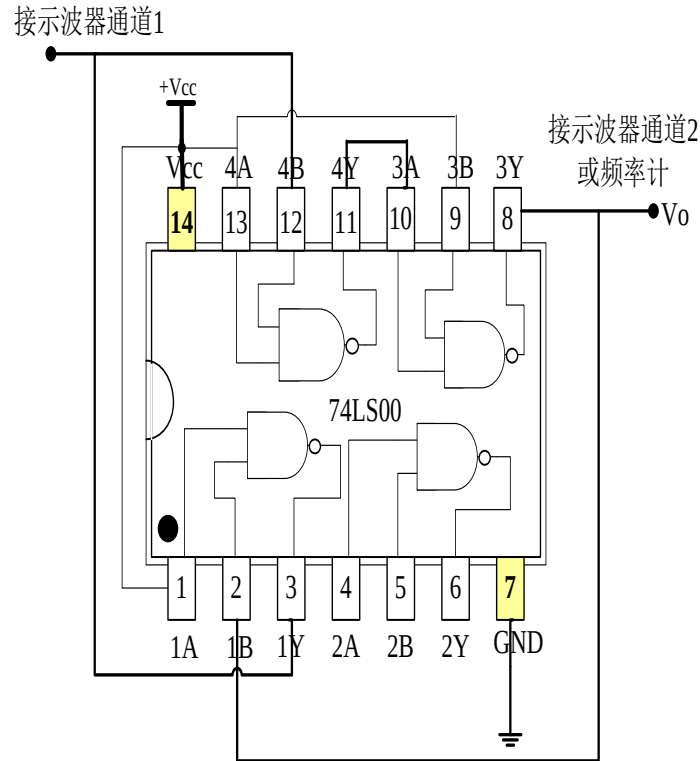


Figure 6 Measurement of transmission delay time t_{pd} for 74LS00 Logic Gate setup
key to build the circuit: one pin of each NAND gate in 74LS00 is connected to high level, the other pins are connected in series.

- Put the IC 74LS00 into socket of the Experiment Box. Pay attention to the direction of the chip.
- Connected circuit according to the diagram, port 14 is connected to V_{cc} and port 7 is to ground.
- Connect Oscilloscope to any input or output of the oscillator
- Adjust the frequency to get the waveform of V_o , obtain the period T and calculate t_{pd}

4.4 Measure t_{pd} for CD4001.

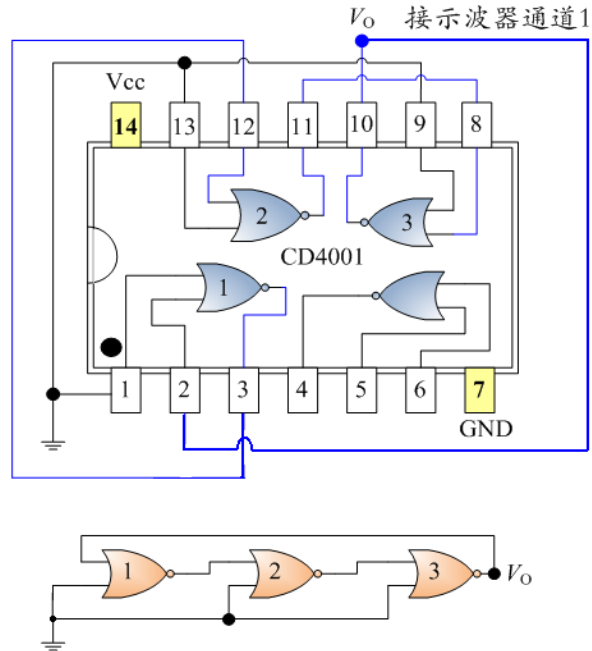


Figure 7 Measurement of transmission delay time t_{pd} for 74LS00 Logic Gate setup
key to build the circuit: one pin of each NAND gate in CD4001 is connected to ground, the other pins are connected in series.

- Put the IC CD4001 into socket of the Experiment Box. Pay attention to the direction of the chip.
- Connected circuit according to the diagram, port 14 is connected to V_{cc} and port 7 is to ground.
- Connect Oscilloscope to any input or output of the oscillator
- Adjust frequency to get the waveform of V_O , obtain the period T and calculate t_{pd}

4.5 Measure the voltage transfer characteristic and on-level and off-level V_{ON} and V_{OFF} for 74LS00.

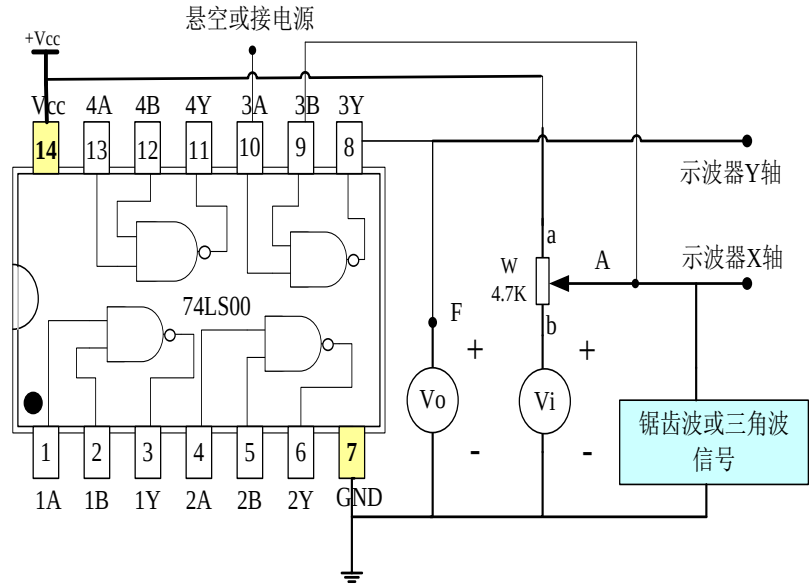


Figure 8 Measurement of 74LS00 Transmission characteristics and Open / close level V_{ON} and V_{OF}

- Put the 74LS00 IC into socket of the Experiment Box.
- Connect circuit according to figure 8.
- Connect ampere meter to input A and output of 2Y.
- Slowly adjust the potentiometer W from b-side to a-side, observe the reading of two voltmeter V_i , V_o and record the data into the table.
- According to the table, draw the graph, and calculate V_{ON} and V_{OFF}

V. Experiment Result and Analysis

5.1 Verify logic function of NAND gate with 74LS00.

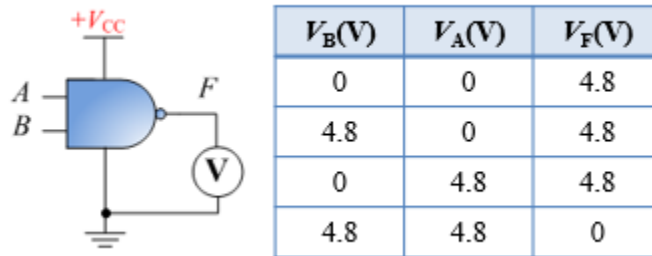


Figure 9 Truth table for NAND gate

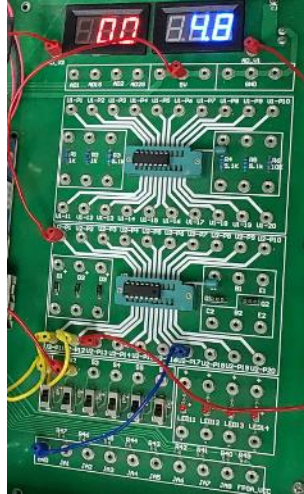


Figure 10 74LS00 connection

5.2 Verify logic function of NOR gate with CD4001.

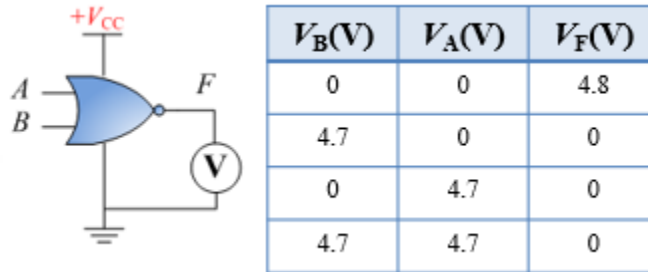


Figure 11 NOR gate truth table

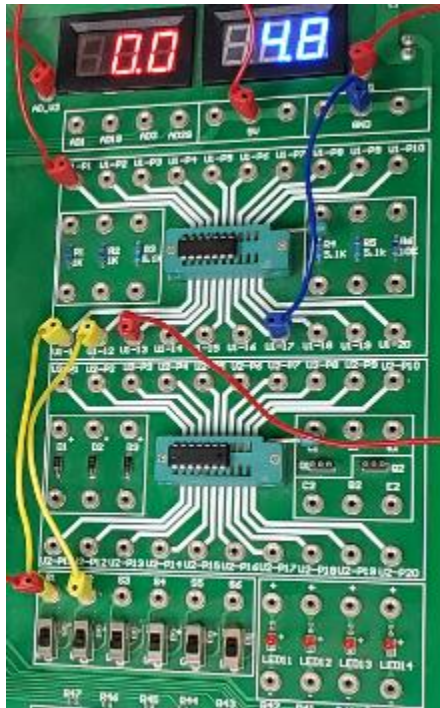


Figure 12 CD4001 connection

5.3 Measure t_{pd} for 74LS00.



Figure 13 Oscilloscope reading
8.4V and 100ns

5.4 Measure t_{pd} for CD4001.



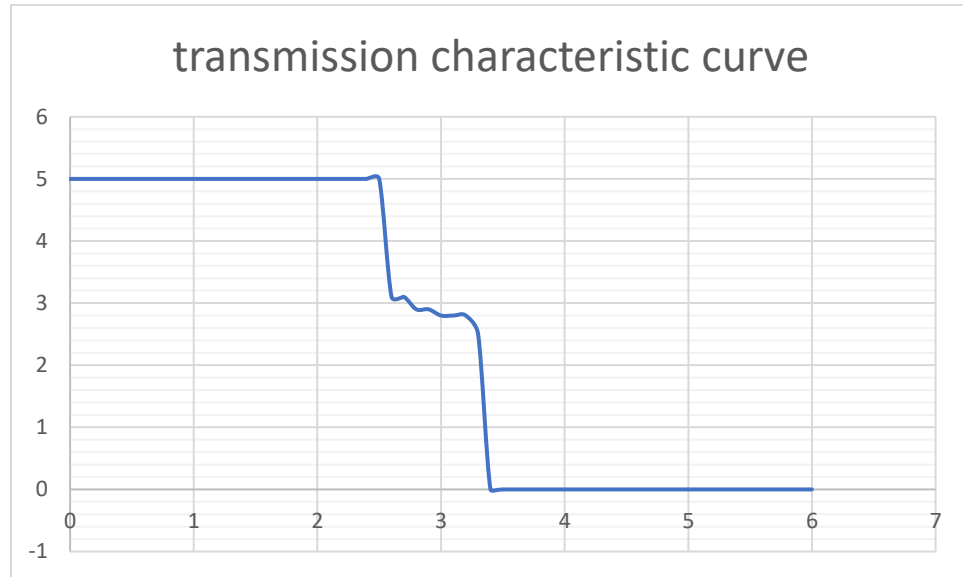
Figure 14 Oscilloscope reading
4V and 300ns

5.5 Measure the voltage transfer characteristic and on-level and off-level V_{ON} and V_{OFF} for 74LS00.

Figure 15 Table for V_{ON}/V_{OFF}

Vi	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8
V	5	5	5	5	5	5	5	5	5
Vi	1.0	1.2	1.4	1.6	1.8	2	2.1	2.2	2.3
V	5	5	5	5	5	5	5	5	5
Vi	2.4	2.5	2.6	2.7	2.8	2.9	3.0	3.1	3.2

V	5	5	3.1	3.1	2.9	2.9	2.8	2.8	2.8
V _i	3.3	3.4	3.5	3.6					
V	2.5	0	0	0					



Vertical represents V
Horizontal represents V_i

Figure 16 74LS—transmission characteristic curve

$V_{ON}=2.6V$ $V_{OFF}=3.4V$

VI. Discussion and Revision

In the experiment propagation delay was introduced, it is an important concept to take note of when building logic circuits. Got to know the characteristics of 74LS00NAND and CD4001NOR logic chips. During the experiment, we had trouble measuring the transmission characteristic. With the instructor's help, we diagnosed poor wire connection.

Lab 4 -- Introduction to Electronic Instruments Lab Report

Name: Chen Yi Hui	ID: 3180300684	Major: Computer Science
Course: Digital Logic Design	Groupmates: Tang Anna, 伍美萍, Choi Justin, Malmaerg Joshua, 冯乐人	
Date:	Place: ZiJinGang East 4-509	Instructor: Hong QiJun

I. Experiment Purpose and Requirements

- 1.1 Master Verilog HDL and design basic logic component with it, and practice code input, edit, debug simulation and synthesis in Xilinx ISE.
- 1.2 Master SWORD FPGA development platforms, and learn how to set time constraint, pin constraints and wiring mapping and sequential simulation.
- 1.3 Use Xilinx ISE to verify the design, then download the code to the experiment board and verify the function of circuit on the board

II. Experiment Contents and Principles

- 2.1 Design circuit with graphical input
 - Use switches as control of the lamp.
- 2.2 Design circuit with Verilog HDL
 - Use buttons as control of the lamp, the lamp is off after it is on for several seconds.

III. Experiment Instruments and Materials

- | | | |
|-----|------------------------|---|
| 3.1 | Experiment Instruments | |
| | PC with ISE 14.7 | 1 |
| | SWORD Board | 1 |

IV. Experiment Procedures and Operations

- 4.1 Design simple combinational logic circuit with graphic interface: create Xilinx ISE project on FPGA Digital system, then edit, debug, use behavior simulation, set time constraint, package pin and map the. After that do the sequential simulation and download the FPGA program file, run and verify.
 - Open ISE 14.7
 - Create new project and name it LampCtrl_sch

- In the project setting, make sure to follow settings in figure 1 and press “Next”

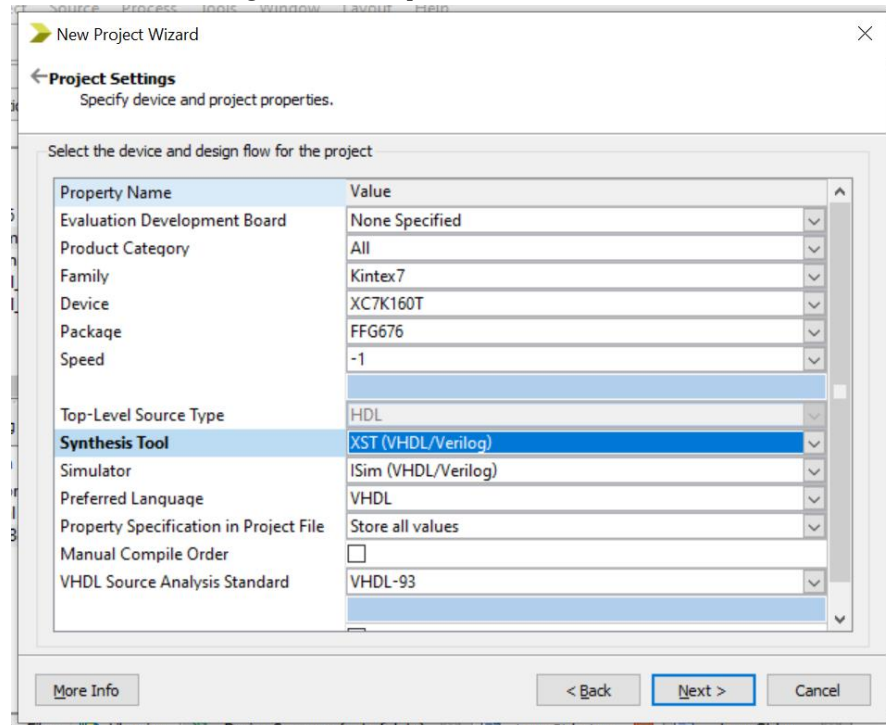


Figure 1 new project setting

- On the “view” section, there would be a file call lampCtrl, right click on it and add “new source”. Select Schematic in the option and name the file LampCtrl.

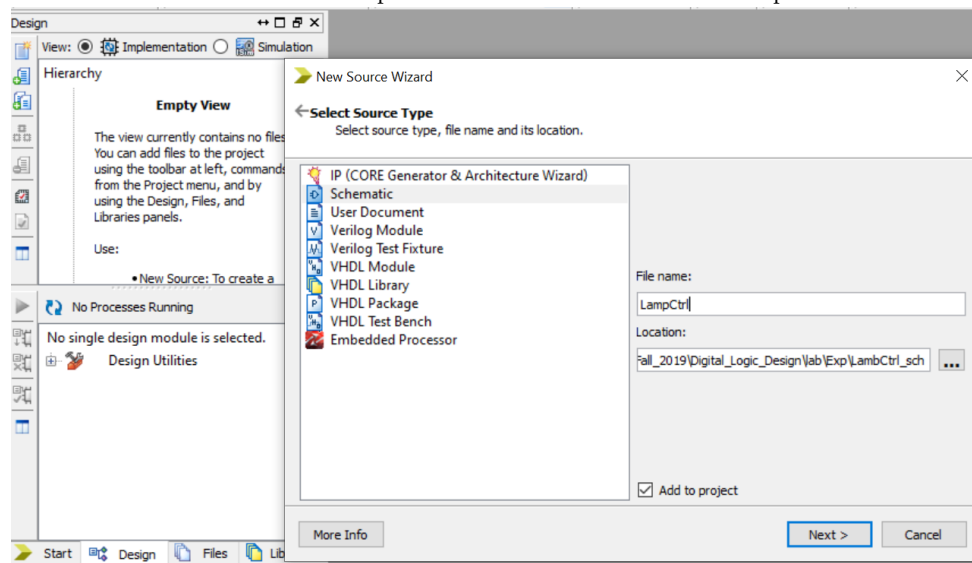


Figure 2 Creating schematic file

- A sketch area will be created, and logic circuit can be drawn on the sketch area. Symbols can be found by searching above red line and search result will be displayed in red circle as shown in figure 3.

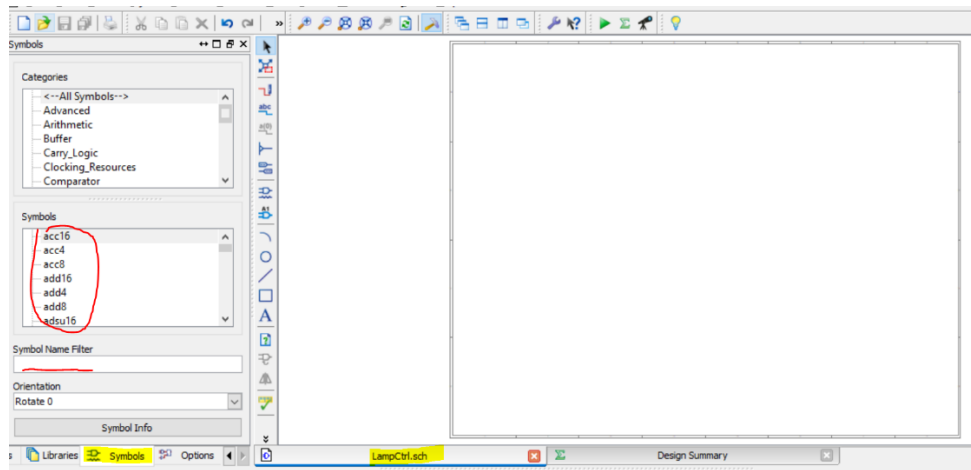


Figure 3 New schematic file created

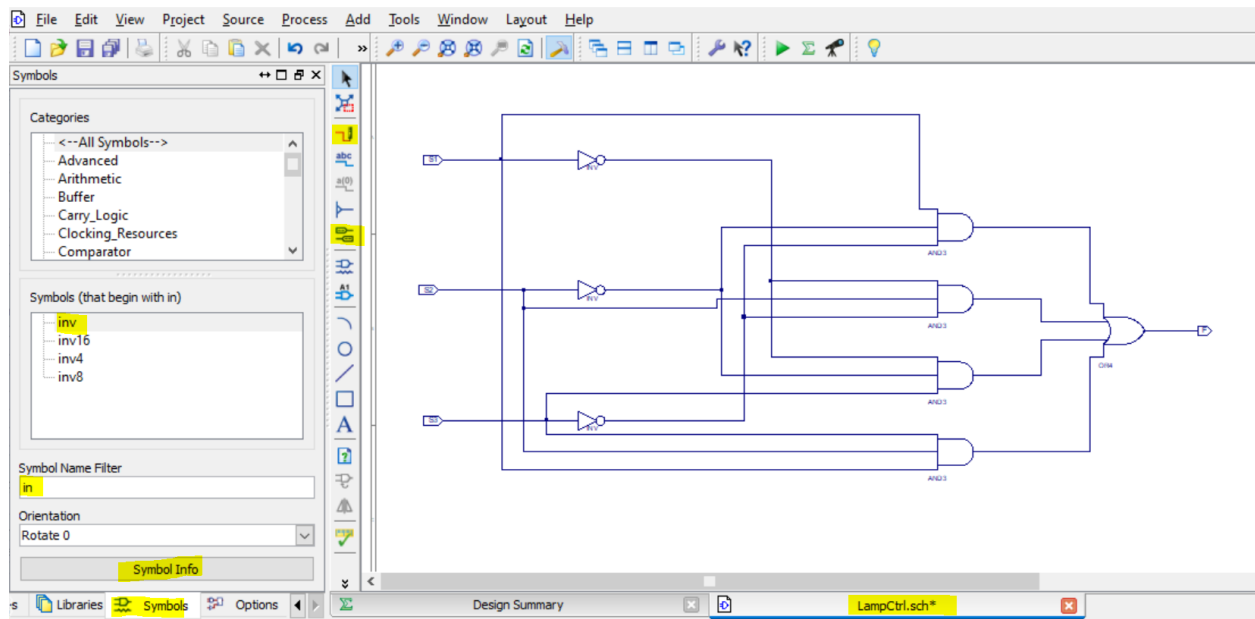


Figure 4 a complete circuit for this experiment

- After a graphing a schematic, click on the schematic file and expand “Design Utilities”, and then double click “Check Design Rules” to check for completed graph.
- Right click schematic file and add “New Source”, select “Verilog Test Fixture” to create a testing file.

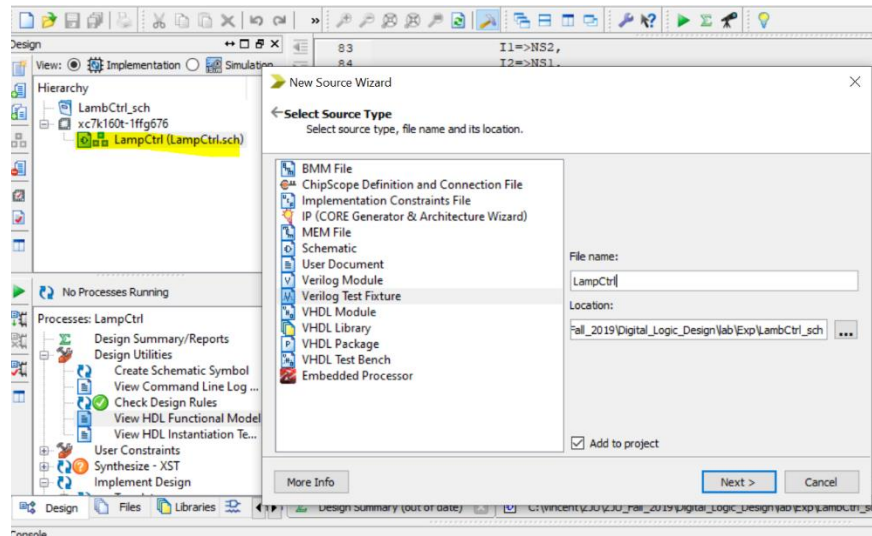


Figure 5 Creating Verilog Test Fixture

- Verilog Test Fixture file can only shows in simulation. In the initial section, type in the following Verilog code:

```
initial begin
S1 = 0;
S2 = 0;
S3 = 0;
S1=1; #50;
S2=1; S1=0; #50;
S1 = 1; #50;
S3 = 1; S2 = 0; S1 = 0; #50;
S1 = 1; #50;
S2 = 1; S1 = 0; #50;
S1 = 1; #50;
end
```

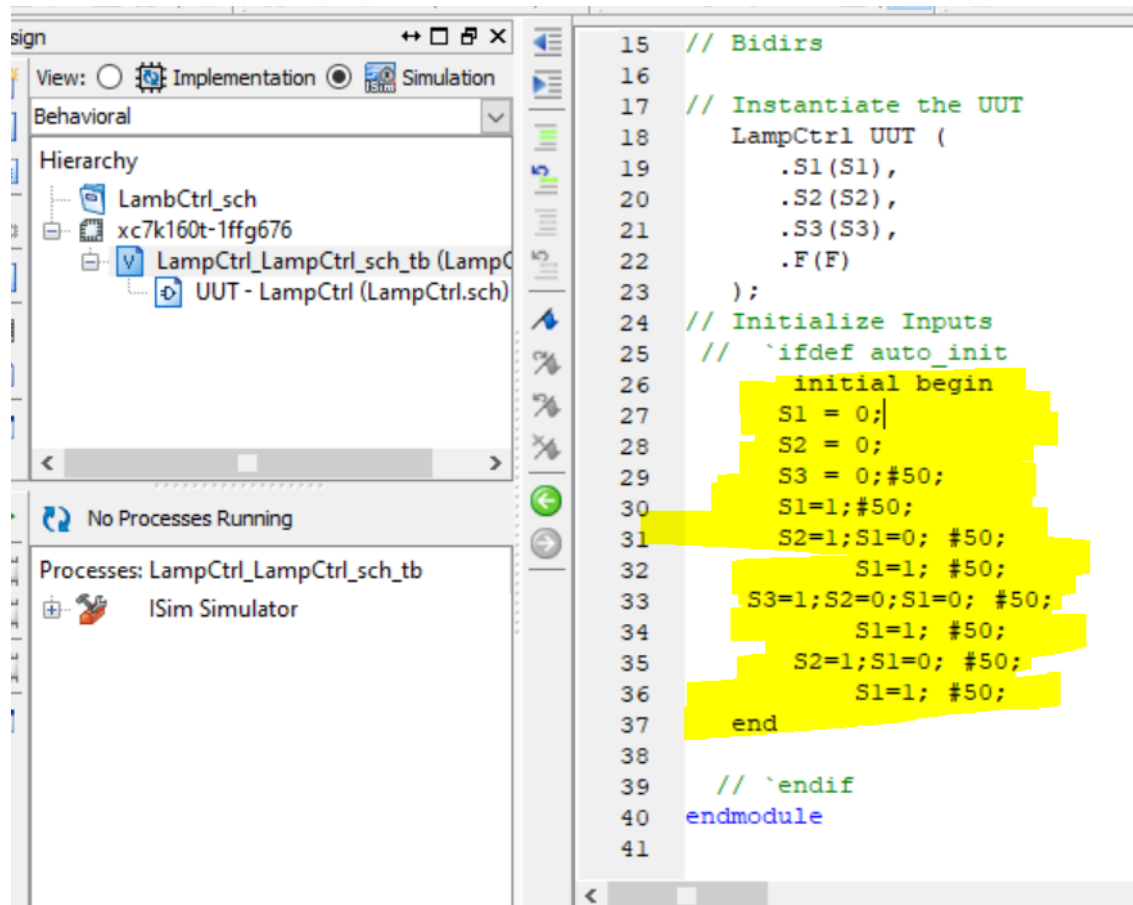


Figure 6 Verilog Test Fixture

- Click on the Verilog Test Fixture double click “Behavior Check Syntax” to check for any syntax error. If no error occurs, double click “Simulate Behavioral Model” to view file model.

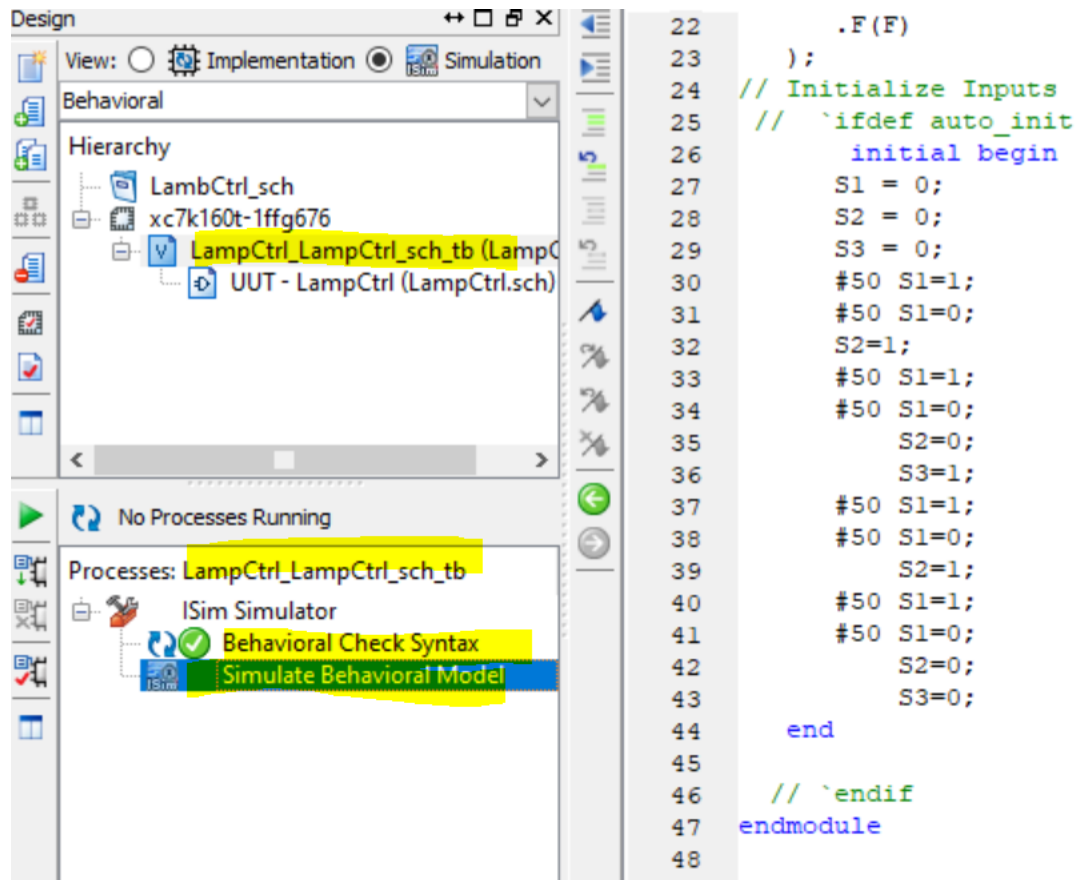


Figure 7 Verilog Test Fixture “Simulate Behavioral Model”

- A model should be displayed, and we can see a graph showing different output according to different combination of inputs.

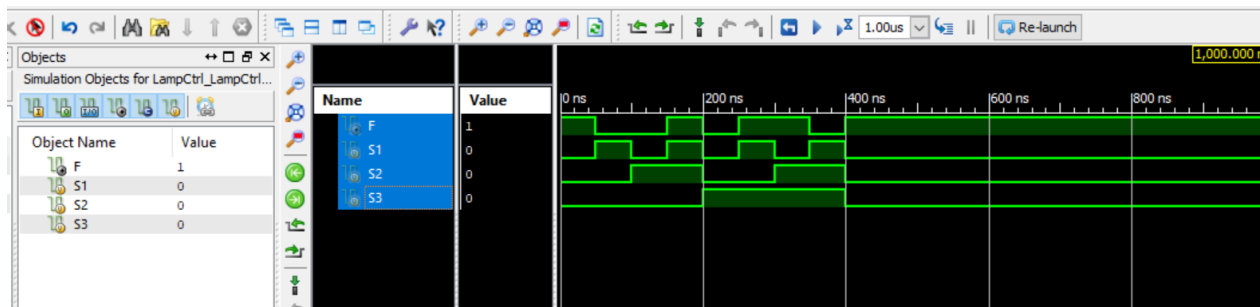


Figure 8 Behavioral Model

- Switch back to “Implementation” window and right click on the lampCtrl file to add “New Source”. Select “Implementation Constraints File”, name it “K7.ucf”.

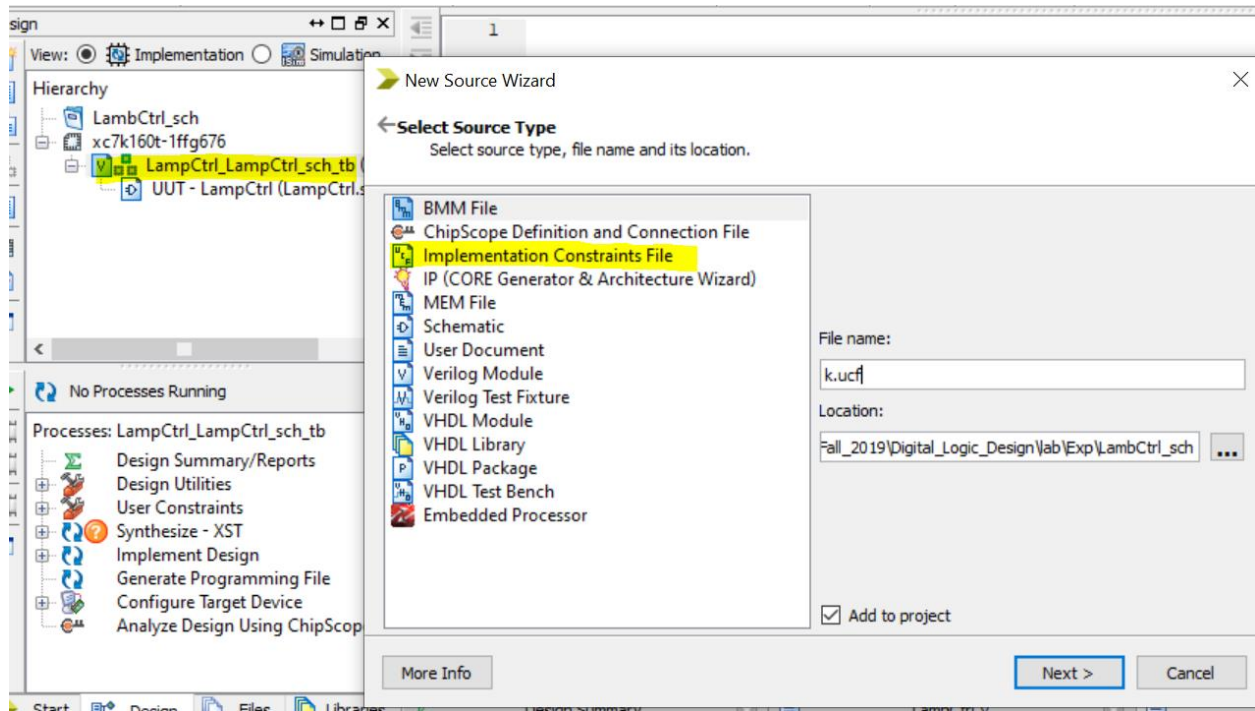


Figure 9 Creating constraints file

- Input the following code:

```
NET "clk" LOC = AC18 | IOSTANDARD=LVCMOS18;
NET "S1" LOC = AA10 | IOSTANDARD=LVCMOS15;
NET "S2" LOC = AB10 | IOSTANDARD=LVCMOS15;
NET "S3" LOC = AA13 | IOSTANDARD=LVCMOS15;
NET "F" LOC = AF24 | IOSTANDARD=LVCMOS33;
```

- Double click "Synthesize" in "Processes" window of main file. Double click "Implement Design", then "Generate Programming File"

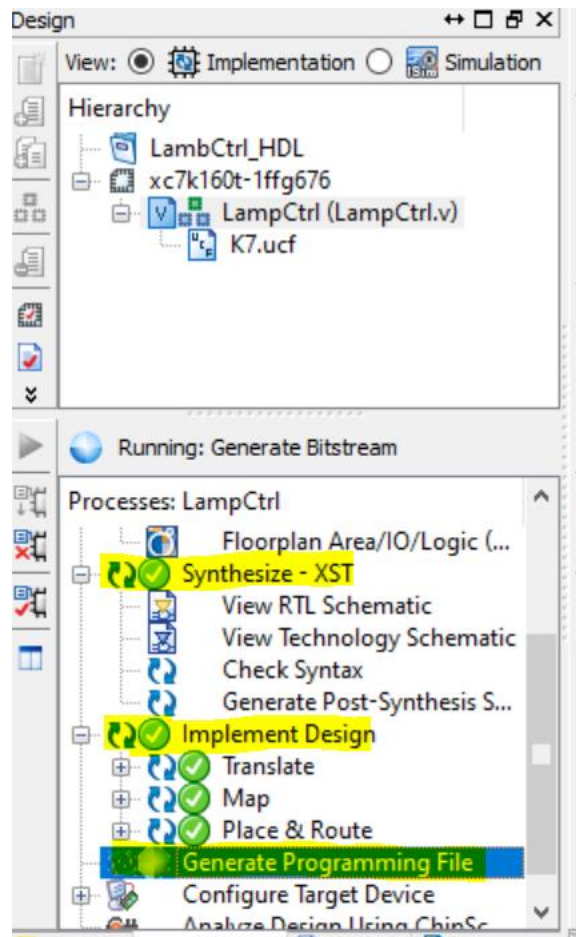


Figure 10 “Processes” window

- Read “Design Summary/Reports” to see design summary

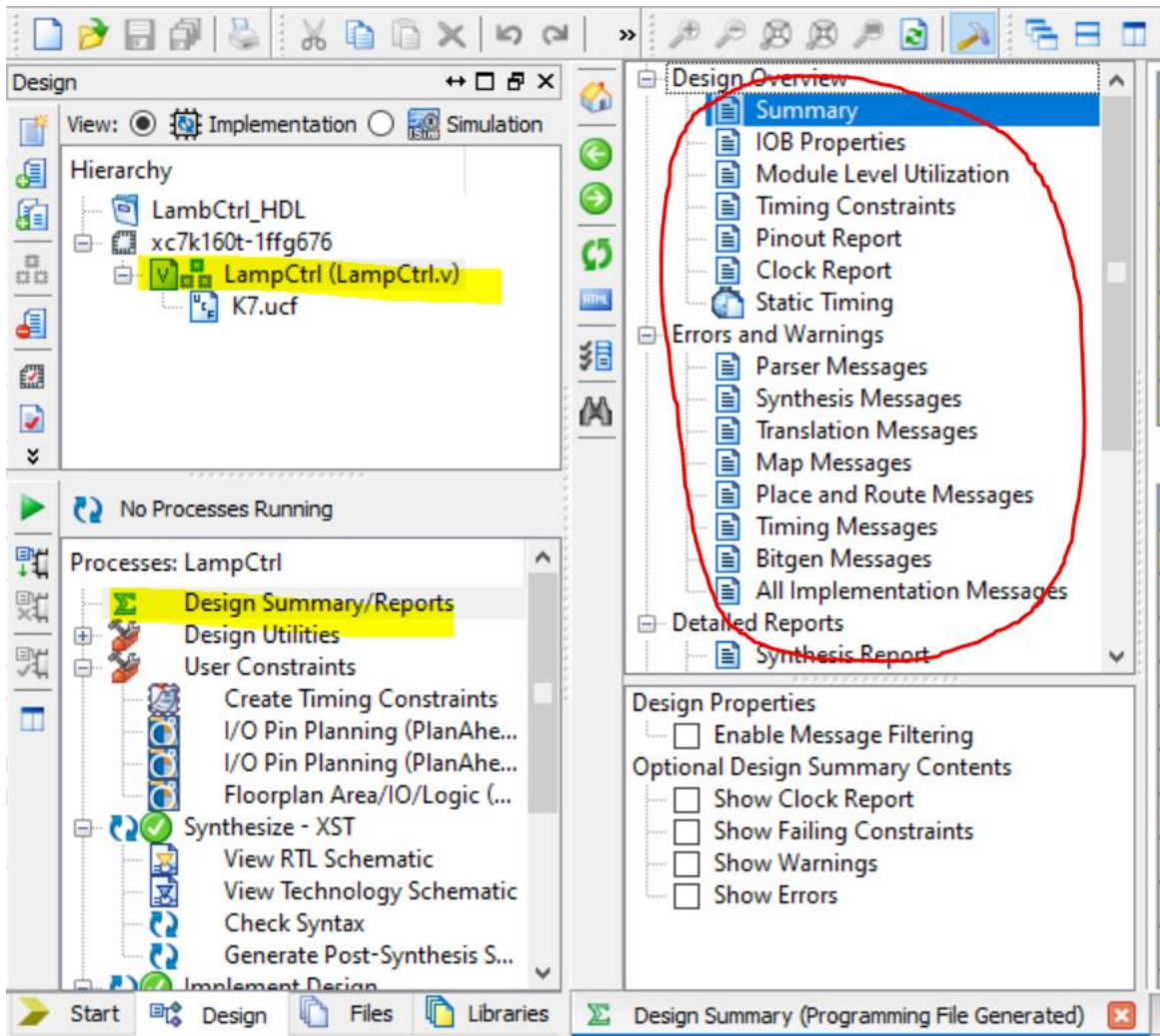
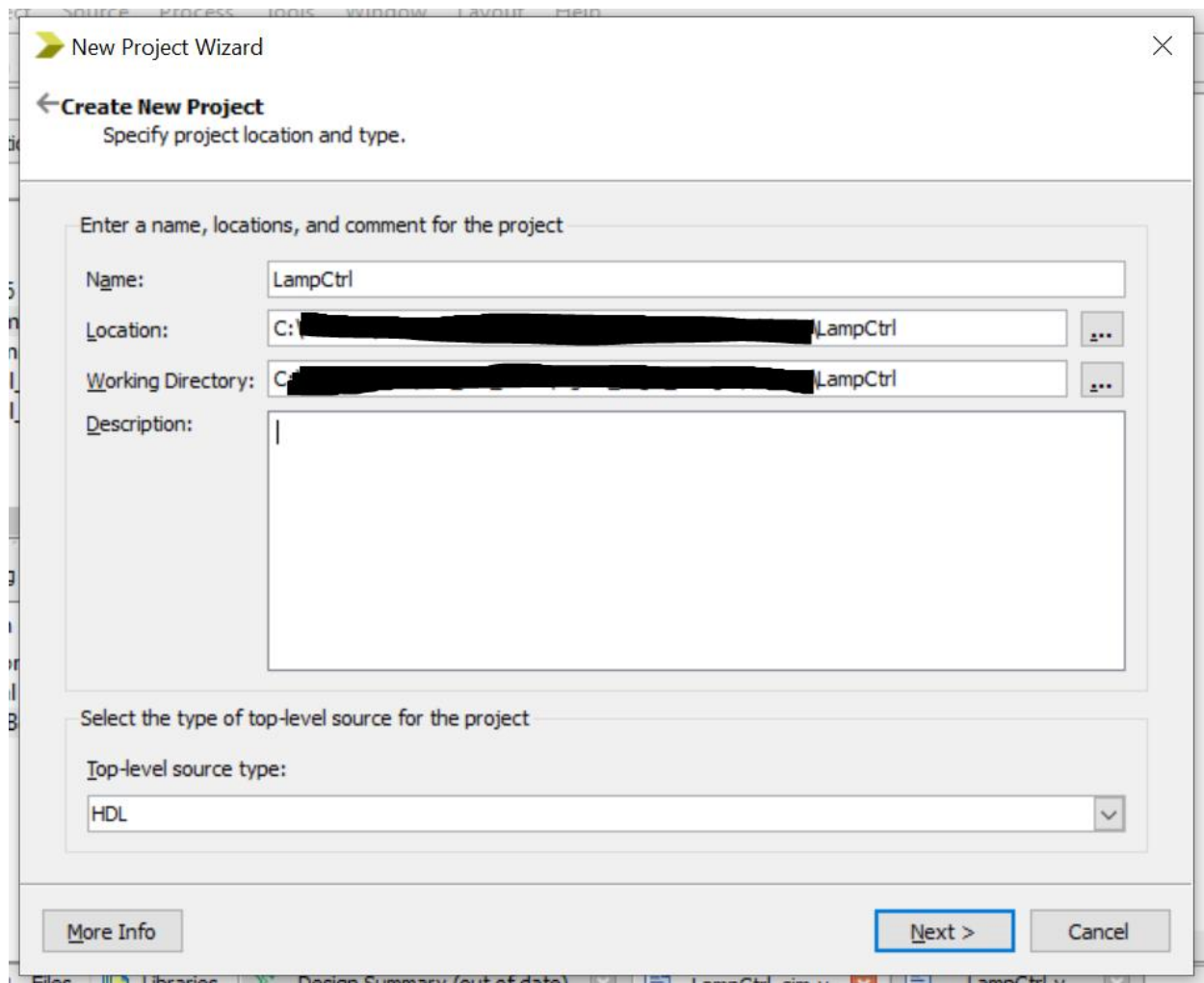


Figure 11 Design Summary/Reports

- If any error occurred, error messages can be read by clicking options in read circle.

4.2 Design simple sequential logic circuit with Verilog code: create Xilinx ISE project on FPGA Digital system, then edit, debug, use behavior simulation, set time constraint, package pin and map the wiring. After that do the sequential simulation and download the FPGA program file, run and verify.

- Create new project



The image shows a 'New Project Wizard' dialog box with a close button (X) in the top right corner. The title bar reads 'New Project Wizard'. Below the title bar, there is a back arrow icon followed by the text 'Create New Project' and a subtitle 'Specify project location and type.'.

The main area of the dialog is divided into two sections. The first section is titled 'Enter a name, locations, and comment for the project'. It contains four input fields: 'Name:' with the text 'LampCtrl', 'Location:' with a path starting 'C:' followed by a redacted path and 'LampCtrl', 'Working Directory:' with a similar redacted path and 'LampCtrl', and 'Description:' which is an empty text area.

The second section is titled 'Select the type of top-level source for the project'. It contains a label 'Top-level source type:' and a dropdown menu currently showing 'HDL'.

At the bottom of the dialog, there are three buttons: 'More Info' on the left, 'Next >' in the center (highlighted with a blue border), and 'Cancel' on the right.

Figure 12 Create new project

- Use proper setting

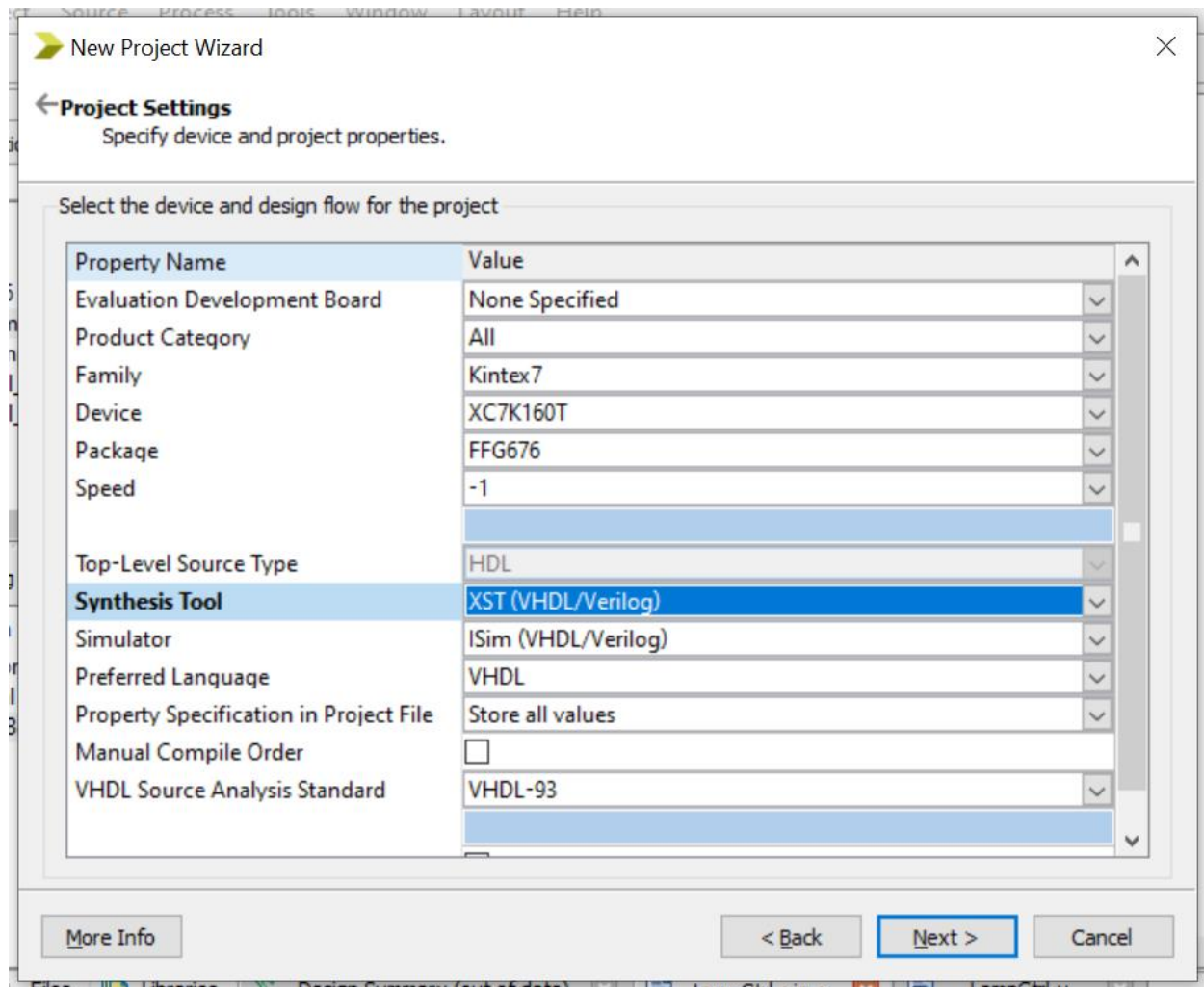


Figure 13 proper setting for project

- On the file created, right click and add “New Source”.

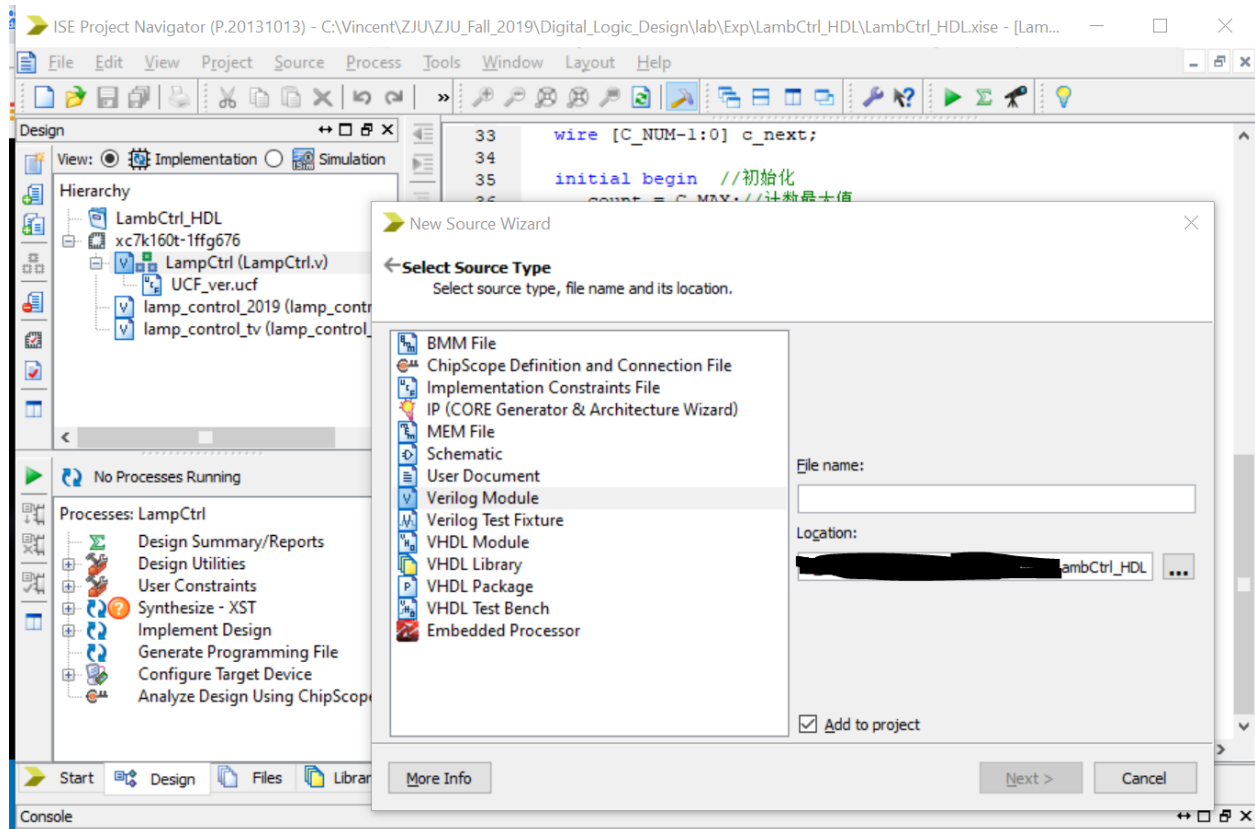


Figure 14 Creating Verilog file

- Implement the Verilog file with following code:

```

module LampCtrl(
    input wire clk,
    input wire S1,
    input wire S2,
    input wire S3,
    output wire F
);

parameter C_NUM = 8;
parameter C_MAX = 8'hFF;

reg [C_NUM-1:0] count;
wire [C_NUM-1:0] c_next;

initial begin//初始化
count = C_MAX;//计数最大值
end
//button pressed
assign w=S1^S2^S3;

//lamp logic
assign F = (count < C_MAX) ? 1'b1 : 1'b0;
//<时F=1;=C_MAX时为0
//count logic
always@(posedge clk)
begin
    if(w == 1'b1)
        count = 0;
    else if(count < C_MAX)
        count = c_next;
end
//next logic
assign c_next = count + 1'b1;

endmodule

```

- In the “Processes” window, click “Synthesize” and double click “Check Syntax”

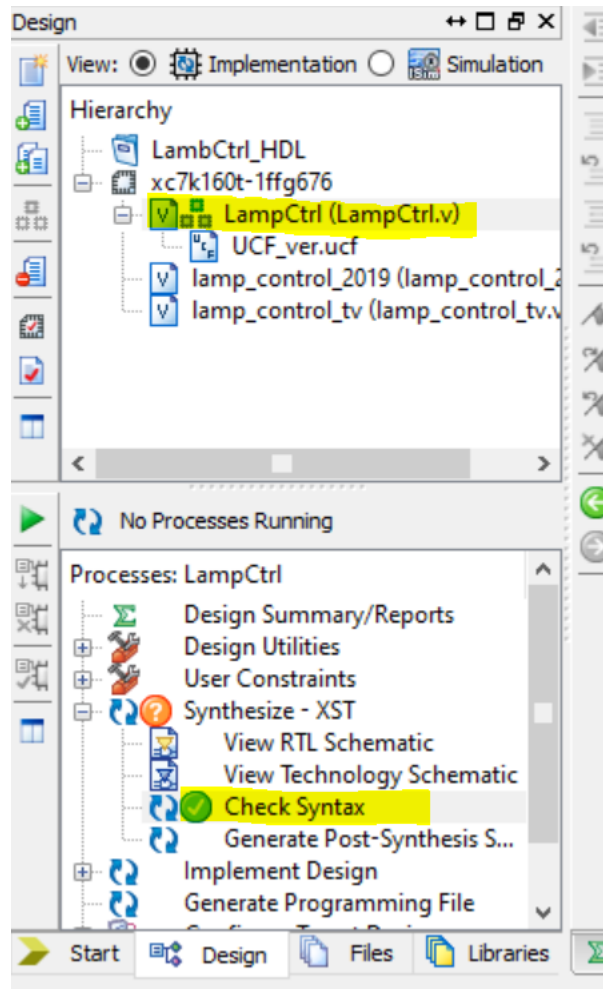


Figure 15 Check Syntax

- Create “Verilog Test Fixture”

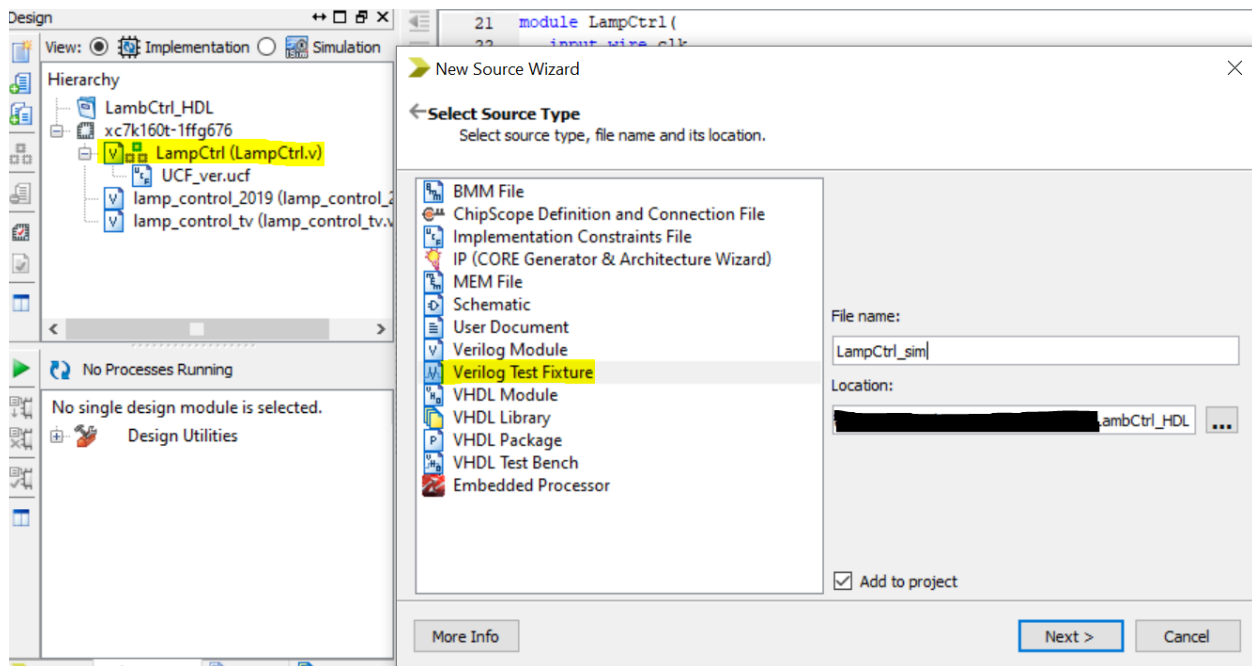


Figure 16 Create Verilog test fixture file

- Find the test file in “Simulation” and implement initial section with following code:

```
module lamp_ctrl_sim;
    // Inputs
    reg clk;
    reg S1;
    reg S2;
    reg S3;
    // Outputs
    wire F;

    // Instantiate the Unit Under Test (UUT)
    lamp_ctrl uut (
        .clk(clk),
        .S1(S1),
        .S2(S2),
        .S3(S3),
        .F(F)
    );

    initial begin
        // Initialize Inputs
        clk = 0;
        S1 = 0; S2 = 0; S3 = 0;

        #600 S1 = 1;
        #20 S1 = 0;
        #6000 S2 = 1;
```

```

        #20 S2 = 0;
        #6000 S3 = 1;
        #20 S3 = 0;

    end

    always begin
        #10 clk = 0;
        #10 clk = 1;

    end
endmodule

```

- Run “Behavioral Check Syntax” and “Simulate Behavioral Model”
-

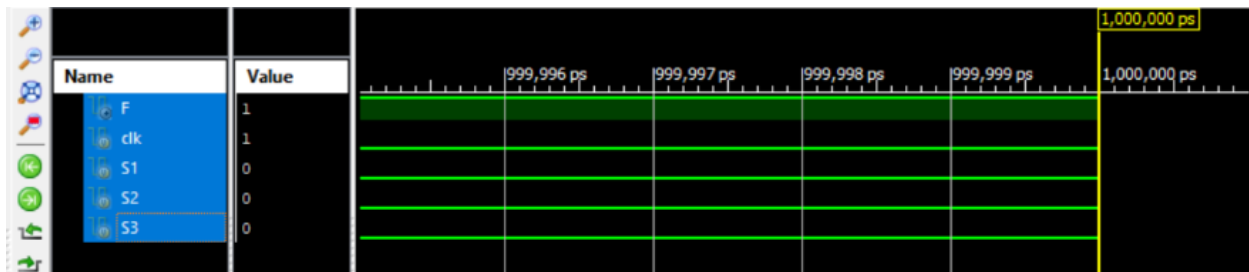


Figure 17 Behavioral model in default

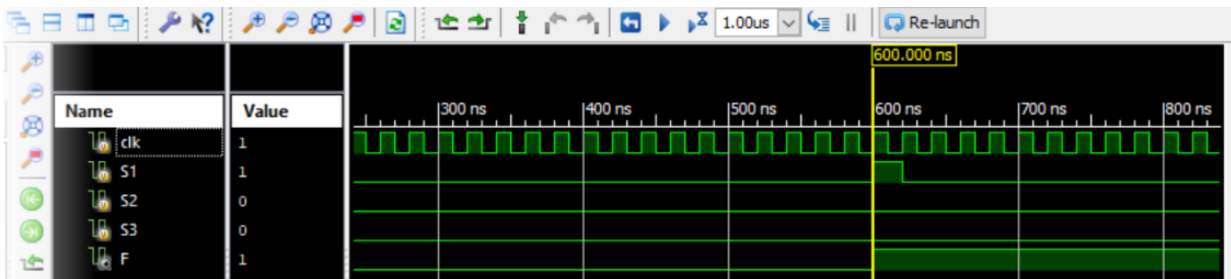
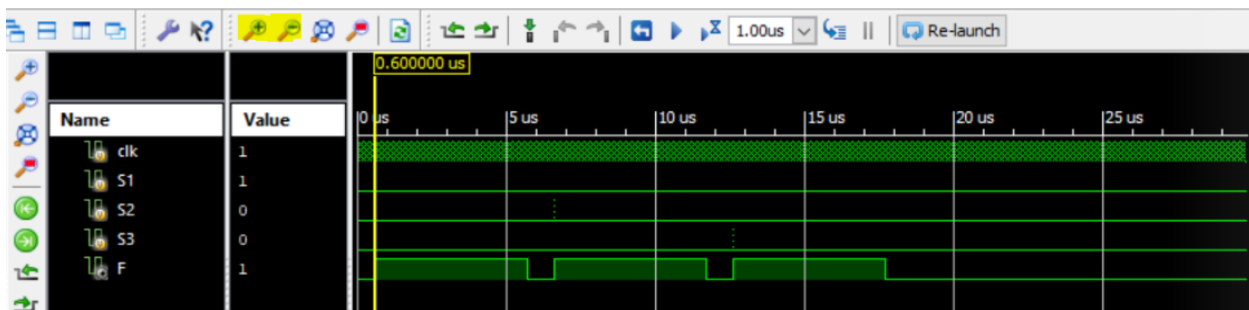


Figure 18 Behavioral model when expanded



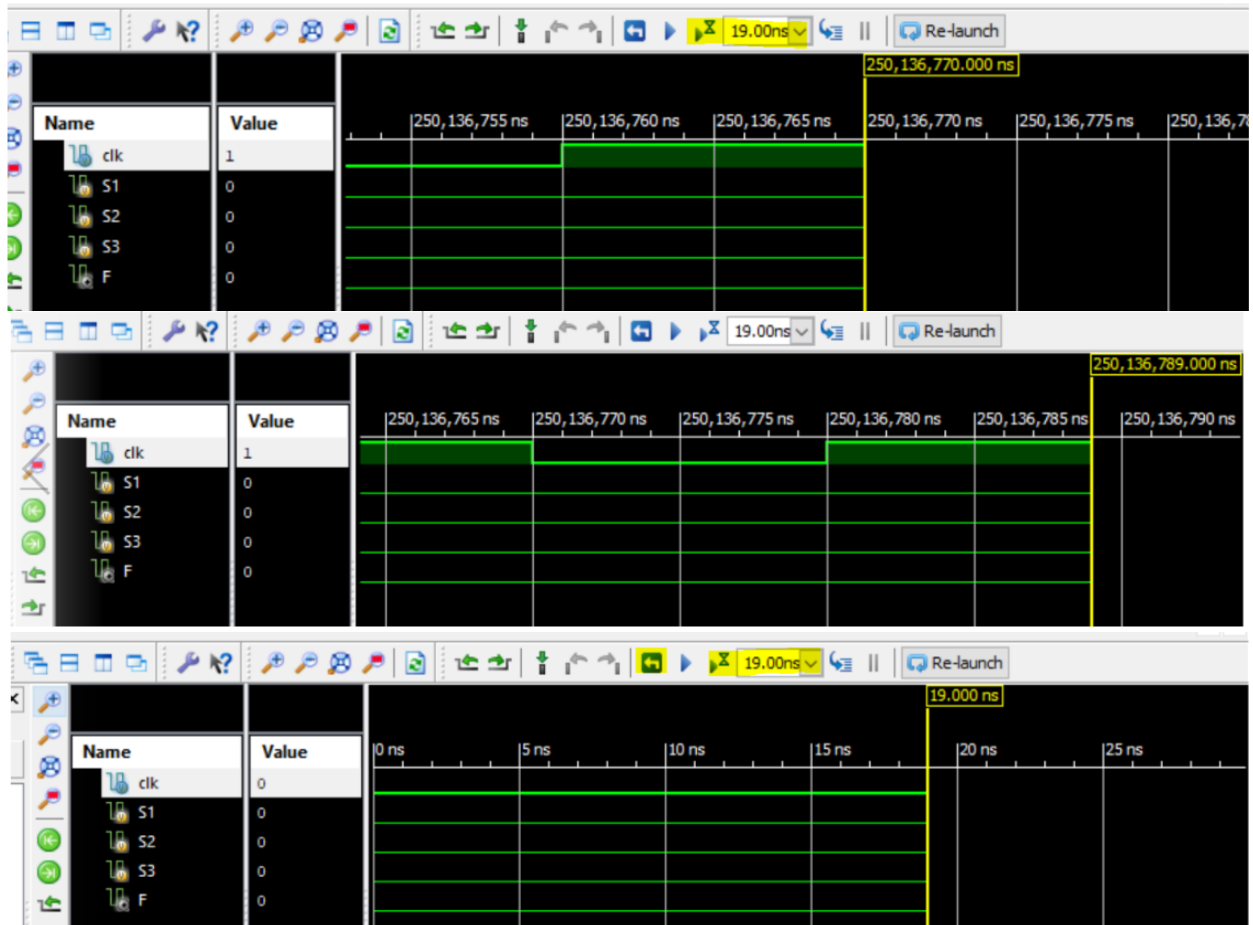


Figure 19-22 Same model with adjusted view settings

- Back to “Implementation” window and add “Implementation Constraints File” to the main project file. Implement the file with following code:

```
NET "clk"LOC = AC18 | IOSTANDARD=LVCMOS18;
NET "S1"LOC = AA10 | IOSTANDARD=LVCMOS15;
NET "S2"LOC = AB10 | IOSTANDARD=LVCMOS15;
NET "S3"LOC = AA13 | IOSTANDARD=LVCMOS15;
NET "F"LOC = AF24 | IOSTANDARD=LVCMOS33;
```

- Run “Synthesize”, “Implement Design” and “Generate Program File”
- Download the program file to SWORD board and run tests.

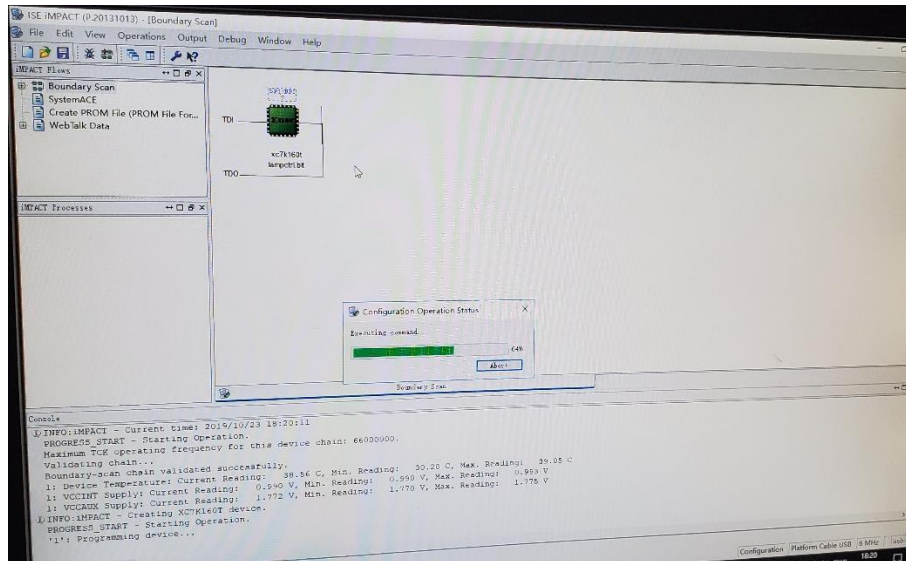


Figure 23 Downloading program file

V. Experiment Result and Analysis

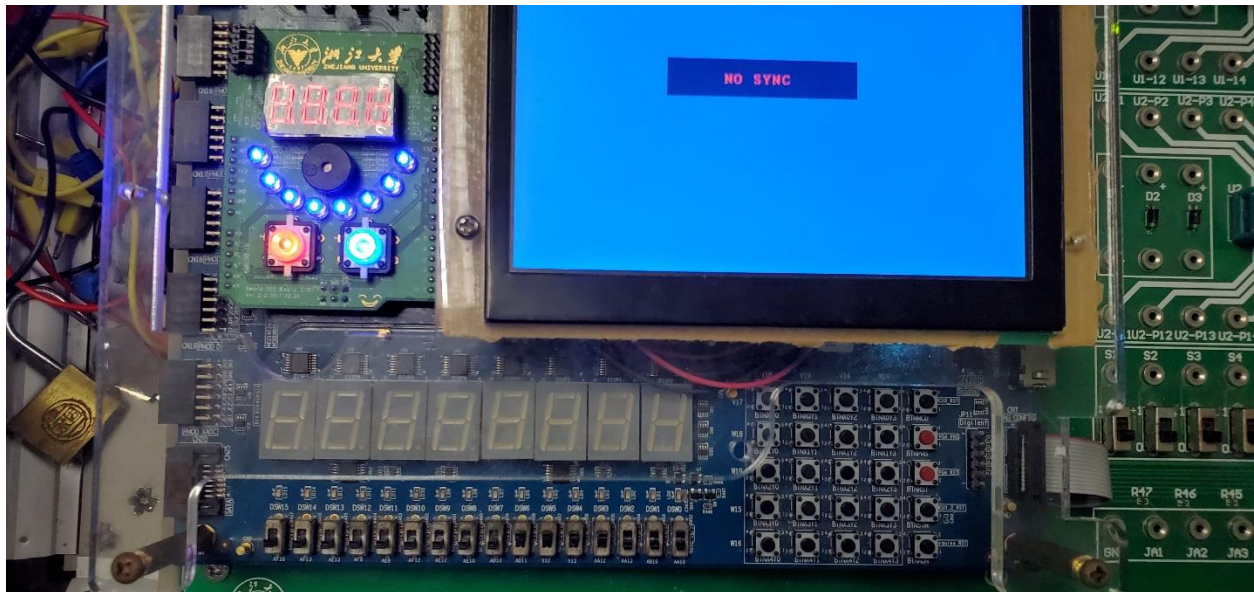


Diagram 24 Result



Diagram 25 Result

VI. Discussion and Revision

Logic graph is a very convenient way to design logic circuits. Procedures might be hard to follow if not operating on the right object, confusing if lack of experience when working on ISE software. For example, Verilog test file only contains in “Simulation”. And when implementing a “Implementation Constraints File” copying and pasting could cause format error. Instructor suggested to type everything manually or plate implementation in a txt file and then copy formatted code in txt file and paste code in the ucf file. Also, I Learnt that there are two ways to formulate a circuit. One way is to graph it, and the other is to code it in Verilog language. I find it easier to graph a logic circuit because it is more visible, and it is easy to follow very wire in the graph.

